WithCoxExport

April 15, 2026

1 LendingClub discrete-time hazard model project

2 File: data/raw/accepted_2007_to_2018Q4.csv.gz

3

4 Workflow:

5 1. Load and clean data

6 2. Define event and censoring

7 3. Create duration

8 4. Expand to loan-month panel

9 5. Train/valid/test split

10 6. Logistic hazard model

11 7. Random forest comparison

12 8. Cumulative PD

13 9. Expected loss / reserve / pricing summaries

[135]:
```
import warnings
warnings.filterwarnings("ignore")

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from pathlib import Path

from sklearn.compose import ColumnTransformer
```

```python
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, average_precision_score

pd.set_option("display.max_columns", 200)
pd.set_option("display.max_rows", 100)
```

[136]:
```python
DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
PROCESSED_DIR = DATA_DIR / "processed"
PROCESSED_DIR.mkdir(parents=True, exist_ok=True)

accepted_path = RAW_DIR / "lendingclub" /"accepted_2007_to_2018Q4.csv.gz"

# Runtime controls
USE_SAMPLE = True          # set to False if you want the full dataset
SAMPLE_SIZE = 90000        # increase later if your machine can handle it
RANDOM_STATE = 42
```

[137]:
```python
df_raw = pd.read_csv(
    accepted_path,
    low_memory=False,
    compression="gzip"
)

print("Raw shape:", df_raw.shape)
print("First 20 columns:", df_raw.columns.tolist()[:20])
```

```
Raw shape: (2260701, 151)
First 20 columns: ['id', 'member_id', 'loan_amnt', 'funded_amnt',
'funded_amnt_inv', 'term', 'int_rate', 'installment', 'grade', 'sub_grade',
'emp_title', 'emp_length', 'home_ownership', 'annual_inc',
'verification_status', 'issue_d', 'loan_status', 'pymnt_plan', 'url', 'desc']
```

[138]:
```python
key_columns = [
    "id",
    "issue_d",
    "loan_status",
    "term",
    "loan_amnt",
    "installment",
    "emp_length",
    "home_ownership",
    "annual_inc",
    "verification_status",
```

```
    "purpose",
    "addr_state",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_range_low",
    "fico_range_high",
    "earliest_cr_line",
    "last_pymnt_d",
    "last_credit_pull_d",
]

missing_cols = [c for c in key_columns if c not in df_raw.columns]
print("Missing columns:", missing_cols)
```

```
Missing columns: []
```

[139]:
```
cols = [c for c in key_columns if c in df_raw.columns]
df = df_raw[cols].copy()

print("Working shape:", df.shape)
df.head()
```

```
Working shape: (2260701, 27)
```

[139]:
```
         id   issue_d loan_status        term  loan_amnt  installment  \
0  68407277  Dec-2015  Fully Paid   36 months     3600.0       123.03
1  68355089  Dec-2015  Fully Paid   36 months    24700.0       820.28
2  68341763  Dec-2015  Fully Paid   60 months    20000.0       432.66
3  66310712  Dec-2015     Current   60 months    35000.0       829.90
4  68476807  Dec-2015  Fully Paid   60 months    10400.0       289.91

  emp_length home_ownership  annual_inc verification_status  \
0  10+ years       MORTGAGE     55000.0        Not Verified
1  10+ years       MORTGAGE     65000.0        Not Verified
2  10+ years       MORTGAGE     63000.0        Not Verified
3  10+ years       MORTGAGE    110000.0     Source Verified
4    3 years       MORTGAGE    104433.0     Source Verified

              purpose addr_state    dti  delinq_2yrs  inq_last_6mths  \
0  debt_consolidation         PA   5.91          0.0             1.0
```

```
1      small_business         SD  16.06          1.0             4.0
2    home_improvement         IL  10.78          0.0             0.0
3  debt_consolidation         NJ  17.06          0.0             0.0
4      major_purchase         PA  25.37          1.0             3.0

   open_acc  pub_rec  revol_bal  revol_util  total_acc  mort_acc  \
0       7.0      0.0     2765.0        29.7       13.0       1.0
1      22.0      0.0    21470.0        19.2       38.0       4.0
2       6.0      0.0     7869.0        56.2       18.0       5.0
3      13.0      0.0     7802.0        11.6       17.0       1.0
4      12.0      0.0    21929.0        64.5       35.0       6.0

   pub_rec_bankruptcies  fico_range_low  fico_range_high earliest_cr_line  \
0                   0.0           675.0            679.0         Aug-2003
1                   0.0           715.0            719.0         Dec-1999
2                   0.0           695.0            699.0         Aug-2000
3                   0.0           785.0            789.0         Sep-2008
4                   0.0           695.0            699.0         Jun-1998

  last_pymnt_d last_credit_pull_d
0     Jan-2019           Mar-2019
1     Jun-2016           Mar-2019
2     Jun-2017           Mar-2019
3     Feb-2019           Mar-2019
4     Jul-2016           Mar-2018
```

[140]:
```
if USE_SAMPLE:
    df = df.sample(SAMPLE_SIZE, random_state=RANDOM_STATE).copy()

print("Shape after optional sampling:", df.shape)
```

```
Shape after optional sampling: (90000, 27)
```

[141]:
```
date_cols = ["issue_d", "earliest_cr_line", "last_pymnt_d",␣
 ↪"last_credit_pull_d"]

for c in date_cols:
    if c in df.columns:
        df[c] = pd.to_datetime(df[c], format="%b-%Y", errors="coerce")

df["fico_avg"] = (df["fico_range_low"] + df["fico_range_high"]) / 2

df["term_num"] = (
    df["term"]
    .astype(str)
    .str.extract(r"(\d+)")
    .astype(float)
```

```
)

df["credit_history_years"] = (
    (df["issue_d"] - df["earliest_cr_line"]).dt.days / 365.25
)

df[["issue_d", "loan_status", "term", "term_num", "fico_avg",␣
 ↪"credit_history_years"]].head()
```

[141]:

```
           issue_d loan_status        term  term_num  fico_avg  \
392949  2015-02-01     Current   60 months      60.0     737.0
1273506 2014-05-01  Fully Paid   36 months      36.0     687.0
324024  2015-04-01  Fully Paid   36 months      36.0     782.0
2066630 2017-12-01  Fully Paid   36 months      36.0     742.0
477199  2018-02-01     Current   60 months      60.0     717.0

         credit_history_years
392949              33.336071
1273506             13.081451
324024              11.581109
2066630             12.501027
477199              18.590007
```

[142]:

```
status_counts = df["loan_status"].value_counts(dropna=False)
status_counts
```

[142]:

```
loan_status
Fully Paid                                          42747
Current                                             34986
Charged Off                                         10805
Late (31-120 days)                                    871
In Grace Period                                       333
Late (16-30 days)                                     173
Does not meet the credit policy. Status:Fully Paid     51
Does not meet the credit policy. Status:Charged Off    30
NaN                                                     4
Name: count, dtype: int64
```

[143]:

```
# Basic event definition:
# event = 1 for default-like outcomes
# event = 0 otherwise

default_statuses = {
    "Charged Off",
    "Default",
}
```

```
df["event"] = df["loan_status"].isin(default_statuses).astype(int)

# Use last payment date if available; otherwise use last credit pull date
df["end_date"] = df["last_pymnt_d"]
mask_missing_end = df["end_date"].isna()
df.loc[mask_missing_end, "end_date"] = df.loc[mask_missing_end,␣
↪"last_credit_pull_d"]

df[["loan_status", "event", "issue_d", "last_pymnt_d", "last_credit_pull_d",␣
↪"end_date"]].head()
```

[143]:

```
        loan_status  event    issue_d last_pymnt_d last_credit_pull_d  \
392949      Current      0 2015-02-01   2019-03-01         2019-03-01
1273506  Fully Paid      0 2014-05-01   2017-06-01         2017-08-01
324024   Fully Paid      0 2015-04-01   2017-03-01         2019-02-01
2066630  Fully Paid      0 2017-12-01   2018-10-01         2018-10-01
477199      Current      0 2018-02-01   2019-03-01         2019-03-01

          end_date
392949  2019-03-01
1273506 2017-06-01
324024  2017-03-01
2066630 2018-10-01
477199  2019-03-01
```

[144]:

```
def month_diff(start, end):
    return (end.dt.year - start.dt.year) * 12 + (end.dt.month - start.dt.month)

df = df[df["issue_d"].notna()].copy()
df = df[df["end_date"].notna()].copy()

df["duration_months"] = month_diff(df["issue_d"], df["end_date"])
df["duration_months"] = df["duration_months"].clip(lower=1)

print(df.shape)
df[["id", "issue_d", "end_date", "duration_months", "event"]].head()
```

```
(89996, 33)
```

[144]:

	id	issue_d	end_date	duration_months	event
392949	39651438	2015-02-01	2019-03-01	49	0
1273506	16411620	2014-05-01	2017-06-01	37	0
324024	45122316	2015-04-01	2017-03-01	23	0
2066630	125356772	2017-12-01	2018-10-01	10	0
477199	128490686	2018-02-01	2019-03-01	13	0

[145]:
```
df = df[df["term_num"].isin([36, 60])].copy()
df = df[df["duration_months"] >= 1].copy()
df = df[df["fico_avg"].notna()].copy()

print("After cleaning/filtering:", df.shape)
```

```
After cleaning/filtering: (89996, 33)
```

[146]:
```
print("Event rate:")
print(df["event"].value_counts(normalize=True))

print("\nLoan status counts:")
print(df["loan_status"].value_counts(dropna=False).head(20))

print("\nDuration summary:")
print(df["duration_months"].describe())
```

```
Event rate:
event
0    0.879939
1    0.120061
Name: proportion, dtype: float64

Loan status counts:
loan_status
Fully Paid                                             42747
Current                                                34986
Charged Off                                            10805
Late (31-120 days)                                       871
In Grace Period                                          333
Late (16-30 days)                                        173
Does not meet the credit policy. Status:Fully Paid        51
Does not meet the credit policy. Status:Charged Off       30
Name: count, dtype: int64

Duration summary:
count    89996.000000
mean        19.570559
std         12.397257
min          1.000000
25%          9.000000
50%         17.000000
75%         29.000000
max         94.000000
Name: duration_months, dtype: float64
```

[147]:
```
loan_level_path = PROCESSED_DIR / "loan_level_survival_base.csv"
df.to_csv(loan_level_path, index=False)
```

```
print(f"Saved loan-level base to: {loan_level_path}")
```

```
Saved loan-level base to: data/processed/loan_level_survival_base.csv
```

[148]:
```
static_cols = [
    "loan_amnt",
    "term_num",
    "installment",
    "emp_length",
    "home_ownership",
    "annual_inc",
    "verification_status",
    "purpose",
    "addr_state",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
]

def expand_loan(row):
    n = int(row["duration_months"])
    event = int(row["event"])

    out = pd.DataFrame({
        "id": row["id"],
        "issue_d": row["issue_d"],
        "month_on_book": np.arange(1, n + 1),
        "duration_months": row["duration_months"],
        "event": row["event"],
    })

    out["hazard_target"] = 0
    if event == 1:
        out.loc[out["month_on_book"] == n, "hazard_target"] = 1

    for col in static_cols:
        out[col] = row[col]
```

```
    return out
```

[149]:
```
panel_parts = []
for _, row in df.iterrows():
    panel_parts.append(expand_loan(row))

panel = pd.concat(panel_parts, ignore_index=True)

print("Panel shape:", panel.shape)
panel.head()
```

```
Panel shape: (1761272, 27)
```

[149]:
```
         id    issue_d  month_on_book  duration_months  event  hazard_target  \
0  39651438 2015-02-01              1               49      0              0
1  39651438 2015-02-01              2               49      0              0
2  39651438 2015-02-01              3               49      0              0
3  39651438 2015-02-01              4               49      0              0
4  39651438 2015-02-01              5               49      0              0

   loan_amnt  term_num  installment emp_length home_ownership  annual_inc  \
0    32000.0      60.0       687.65  10+ years       MORTGAGE    120000.0
1    32000.0      60.0       687.65  10+ years       MORTGAGE    120000.0
2    32000.0      60.0       687.65  10+ years       MORTGAGE    120000.0
3    32000.0      60.0       687.65  10+ years       MORTGAGE    120000.0
4    32000.0      60.0       687.65  10+ years       MORTGAGE    120000.0

  verification_status             purpose addr_state    dti  delinq_2yrs  \
0            Verified  debt_consolidation         CA  24.05          0.0
1            Verified  debt_consolidation         CA  24.05          0.0
2            Verified  debt_consolidation         CA  24.05          0.0
3            Verified  debt_consolidation         CA  24.05          0.0
4            Verified  debt_consolidation         CA  24.05          0.0

   inq_last_6mths  open_acc  pub_rec  revol_bal  revol_util  total_acc  \
0             0.0      20.0      0.0    39687.0        57.8       42.0
1             0.0      20.0      0.0    39687.0        57.8       42.0
2             0.0      20.0      0.0    39687.0        57.8       42.0
3             0.0      20.0      0.0    39687.0        57.8       42.0
4             0.0      20.0      0.0    39687.0        57.8       42.0

   mort_acc  pub_rec_bankruptcies  fico_avg  credit_history_years
0       2.0                   0.0     737.0             33.336071
1       2.0                   0.0     737.0             33.336071
2       2.0                   0.0     737.0             33.336071
3       2.0                   0.0     737.0             33.336071
4       2.0                   0.0     737.0             33.336071
```

[150]:
```
panel["mob_str"] = panel["month_on_book"].astype(str)
panel["fico_x_mob"] = panel["fico_avg"] * panel["month_on_book"]
panel["term_x_mob"] = panel["term_num"] * panel["month_on_book"]
panel["dti_x_mob"] = panel["dti"] * panel["month_on_book"]
```

[151]:
```
# Cleaner time specification
panel["mob_6bin"] = pd.cut(
    panel["month_on_book"],
    bins=[0, 6, 12, 18, 24, 36, 60, 120],
    labels=["1_6", "7_12", "13_18", "19_24", "25_36", "37_60", "61_plus"]
)

panel["month_on_book_sq"] = panel["month_on_book"] ** 2

# optional vintage variable
panel["issue_year"] = panel["issue_d"].dt.year.astype(str)
```

[ ]:
```

```

[152]:
```
panel_path = PROCESSED_DIR / "loan_month_panel.csv"
panel.to_csv(panel_path, index=False)
print(f"Saved panel to: {panel_path}")
```

```
Saved panel to: data/processed/loan_month_panel.csv
```

[153]:
```
train_panel = panel[panel["issue_d"] < "2016-01-01"].copy()
valid_panel = panel[(panel["issue_d"] >= "2016-01-01") & (panel["issue_d"] <␣
 ↪"2017-01-01")].copy()
test_panel  = panel[panel["issue_d"] >= "2017-01-01"].copy()

print("Train shape:", train_panel.shape)
print("Valid shape:", valid_panel.shape)
print("Test shape :", test_panel.shape)
```

```
Train shape: (932040, 34)
Valid shape: (391399, 34)
Test shape : (437833, 34)
```

[154]:
```
target = "hazard_target"

numeric_features = [
    "loan_amnt",
    "term_num",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
```

```
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "fico_x_mob",
    "term_x_mob",
    "dti_x_mob",
]

categorical_features = [
    "emp_length",
    "home_ownership",
    "verification_status",
    "purpose",
    "addr_state",
    "mob_str",       # full month dummy
    "mob_6bin",      # can keep this too
]
```

[155]:
```
X_train = train_panel[numeric_features + categorical_features]
y_train = train_panel[target]

X_valid = valid_panel[numeric_features + categorical_features]
y_valid = valid_panel[target]

X_test = test_panel[numeric_features + categorical_features]
y_test = test_panel[target]

print(X_train.shape, y_train.shape)
print(X_valid.shape, y_valid.shape)
print(X_test.shape, y_test.shape)
```

```
(932040, 26) (932040,)
(391399, 26) (391399,)
(437833, 26) (437833,)
```

[156]:
```
numeric_transformer = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
])

categorical_transformer = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
```

```
    ("onehot", OneHotEncoder(handle_unknown="ignore")),
])

preprocessor = ColumnTransformer([
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features),
])
```

[157]:
```
hazard_model = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000))
])

hazard_model.fit(X_train, y_train)
```

[157]:
```
Pipeline(steps=[('prep',
                 ColumnTransformer(transformers=[('num',
                                                  Pipeline(steps=[('imputer',
SimpleImputer(strategy='median')),
                                                                  ('scaler',
StandardScaler())]),
                                                  ['loan_amnt', 'term_num',
                                                   'installment', 'annual_inc',
                                                   'dti', 'delinq_2yrs',
                                                   'inq_last_6mths', 'open_acc',
                                                   'pub_rec', 'revol_bal',
                                                   'revol_util', 'total_acc',
                                                   'mort_acc',
                                                   'pub_rec_bankruptcies',
                                                   'fico_avg',
                                                   'credit_history_years',
                                                   'fico_x_mob', 'term_x_mob',
                                                   'dti_x_mob']),
                                                 ('cat',
                                                  Pipeline(steps=[('imputer',
SimpleImputer(strategy='most_frequent')),
                                                                  ('onehot',
OneHotEncoder(handle_unknown='ignore'))]),
                                                  ['emp_length',
                                                   'home_ownership',
                                                   'verification_status',
                                                   'purpose', 'addr_state',
                                                   'mob_str', 'mob_6bin'])])),
                 ('model', LogisticRegression(max_iter=2000))])
```

[158]:
```
valid_pred_hazard = hazard_model.predict_proba(X_valid)[:, 1]
test_pred_hazard = hazard_model.predict_proba(X_test)[:, 1]
```

```
hazard_results = {
    "valid_roc_auc": roc_auc_score(y_valid, valid_pred_hazard),
    "valid_pr_auc": average_precision_score(y_valid, valid_pred_hazard),
    "test_roc_auc": roc_auc_score(y_test, test_pred_hazard),
    "test_pr_auc": average_precision_score(y_test, test_pred_hazard),
}

hazard_results
```

[158]:
```
{'valid_roc_auc': 0.6644510998975379,
 'valid_pr_auc': 0.013513873187659108,
 'test_roc_auc': 0.6705443367805358,
 'test_pr_auc': 0.008341943712689478}
```

[159]:
```
from sklearn.calibration import CalibratedClassifierCV
```

[160]:
```
rf_base = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        min_samples_leaf=50,
        random_state=RANDOM_STATE,
        n_jobs=-1,
        class_weight="balanced_subsample"
    )),
])

rf_calibrated = CalibratedClassifierCV(
    estimator=rf_base,
    method="sigmoid",
    cv=3
)

rf_calibrated.fit(X_train, y_train)
```

[160]:
```
CalibratedClassifierCV(cv=3,
                       estimator=Pipeline(steps=[('prep',
ColumnTransformer(transformers=[('num',
   Pipeline(steps=[('imputer',
                    SimpleImputer(strategy='median')),
                   ('scaler',
                    StandardScaler())]),
   ['loan_amnt',
    'term_num',
    'installment',
    'annual_inc',
```

```
     'dti',
     'delinq_2yrs',
     'inq_last_6mths',
     'open_acc',
     'pub_rec',
     'revol_bal',
     'revol_util',
     'total_acc',
     'mort_acc',
     '…
    Pipeline(steps=[('imputer',
                     SimpleImputer(strategy='most_frequent')),
                    ('onehot',
                     OneHotEncoder(handle_unknown='ignore'))]),
    ['emp_length',
     'home_ownership',
     'verification_status',
     'purpose',
     'addr_state',
     'mob_str',
     'mob_6bin'])])),
                                                    ('model',
RandomForestClassifier(class_weight='balanced_subsample',
min_samples_leaf=50,
n_estimators=200,
n_jobs=-1,
random_state=42))]))
```

[161]:
```
valid_pred_rf = rf_calibrated.predict_proba(X_valid)[:, 1]
test_pred_rf = rf_calibrated.predict_proba(X_test)[:, 1]

rf_results = {
    "valid_roc_auc": roc_auc_score(y_valid, valid_pred_rf),
    "valid_pr_auc": average_precision_score(y_valid, valid_pred_rf),
    "test_roc_auc": roc_auc_score(y_test, test_pred_rf),
    "test_pr_auc": average_precision_score(y_test, test_pred_rf),
}

rf_results
```

[161]:
```
{'valid_roc_auc': 0.6568373154296456,
 'valid_pr_auc': 0.012907198473836468,
 'test_roc_auc': 0.6652843789467573,
 'test_pr_auc': 0.008311529282920003}
```

[162]:
```
results_df = pd.DataFrame([
    {
```

```
        "model": "logistic_hazard",
        **hazard_results
    },
    {
        "model": "random_forest",
        **rf_results
    }
])

results_df
```

[162]:

```
              model  valid_roc_auc  valid_pr_auc  test_roc_auc  test_pr_auc
0  logistic_hazard       0.664451      0.013514      0.670544     0.008342
1    random_forest       0.656837      0.012907      0.665284     0.008312
```

[163]:

```
test_scored = test_panel.copy()
test_scored["pred_hazard_logit"] = test_pred_hazard
test_scored["pred_hazard_rf"] = test_pred_rf

test_scored = test_scored.sort_values(["id", "month_on_book"]).copy()
test_scored.head()
```

[163]:

```
               id    issue_d  month_on_book  duration_months  event  \
214997  100001365 2017-03-01              1               24      0
214998  100001365 2017-03-01              2               24      0
214999  100001365 2017-03-01              3               24      0
215000  100001365 2017-03-01              4               24      0
215001  100001365 2017-03-01              5               24      0

        hazard_target  loan_amnt  term_num  installment emp_length  \
214997              0    20000.0      36.0       683.46  10+ years
214998              0    20000.0      36.0       683.46  10+ years
214999              0    20000.0      36.0       683.46  10+ years
215000              0    20000.0      36.0       683.46  10+ years
215001              0    20000.0      36.0       683.46  10+ years

       home_ownership  annual_inc verification_status             purpose  \
214997       MORTGAGE     87000.0        Not Verified  debt_consolidation
214998       MORTGAGE     87000.0        Not Verified  debt_consolidation
214999       MORTGAGE     87000.0        Not Verified  debt_consolidation
215000       MORTGAGE     87000.0        Not Verified  debt_consolidation
215001       MORTGAGE     87000.0        Not Verified  debt_consolidation

       addr_state   dti  delinq_2yrs  inq_last_6mths  open_acc  pub_rec  \
214997         MT  25.3          0.0             0.0      12.0      0.0
214998         MT  25.3          0.0             0.0      12.0      0.0
214999         MT  25.3          0.0             0.0      12.0      0.0
```

```
215000         MT  25.3          0.0             0.0      12.0      0.0
215001         MT  25.3          0.0             0.0      12.0      0.0

        revol_bal  revol_util  total_acc  mort_acc  pub_rec_bankruptcies  \
214997    19262.0        79.3       19.0       1.0                   0.0
214998    19262.0        79.3       19.0       1.0                   0.0
214999    19262.0        79.3       19.0       1.0                   0.0
215000    19262.0        79.3       19.0       1.0                   0.0
215001    19262.0        79.3       19.0       1.0                   0.0

        fico_avg  credit_history_years mob_str  fico_x_mob  term_x_mob  \
214997     662.0             12.495551       1       662.0        36.0
214998     662.0             12.495551       2      1324.0        72.0
214999     662.0             12.495551       3      1986.0       108.0
215000     662.0             12.495551       4      2648.0       144.0
215001     662.0             12.495551       5      3310.0       180.0

        dti_x_mob mob_6bin  month_on_book_sq issue_year  pred_hazard_logit  \
214997       25.3      1_6                 1       2017           0.003000
214998       50.6      1_6                 4       2017           0.003259
214999       75.9      1_6                 9       2017           0.003778
215000      101.2      1_6                16       2017           0.003747
215001      126.5      1_6                25       2017           0.003997

        pred_hazard_rf
214997        0.002148
214998        0.002800
214999        0.003252
215000        0.003844
215001        0.004040
```

[164]:
```
test_scored["survival_prob_logit"] = test_scored.
 ↪groupby("id")["pred_hazard_logit"].transform(
    lambda x: (1 - x).cumprod()
)
test_scored["cum_pd_logit"] = 1 - test_scored["survival_prob_logit"]

test_scored["survival_prob_rf"] = test_scored.groupby("id")["pred_hazard_rf"].
 ↪transform(
    lambda x: (1 - x).cumprod()
)
test_scored["cum_pd_rf"] = 1 - test_scored["survival_prob_rf"]

test_scored[[
    "id", "month_on_book", "pred_hazard_logit", "cum_pd_logit",
    "pred_hazard_rf", "cum_pd_rf"
]].head(20)
```

[164]:

	id	month_on_book	pred_hazard_logit	cum_pd_logit	\
214997	100001365	1	0.003000	0.003000	
214998	100001365	2	0.003259	0.006249	
214999	100001365	3	0.003778	0.010003	
215000	100001365	4	0.003747	0.013713	
215001	100001365	5	0.003997	0.017655	
215002	100001365	6	0.004289	0.021869	
215003	100001365	7	0.008148	0.029838	
215004	100001365	8	0.008010	0.037609	
215005	100001365	9	0.008212	0.045513	
215006	100001365	10	0.008351	0.053484	
215007	100001365	11	0.009257	0.062245	
215008	100001365	12	0.008988	0.070674	
215009	100001365	13	0.009295	0.079312	
215010	100001365	14	0.009230	0.087809	
215011	100001365	15	0.008467	0.095533	
215012	100001365	16	0.009395	0.104030	
215013	100001365	17	0.009852	0.112857	
215014	100001365	18	0.008997	0.120839	
215015	100001365	19	0.009238	0.128960	
215016	100001365	20	0.010132	0.137786	

	pred_hazard_rf	cum_pd_rf
214997	0.002148	0.002148
214998	0.002800	0.004942
214999	0.003252	0.008177
215000	0.003844	0.011990
215001	0.004040	0.015982
215002	0.004896	0.020800
215003	0.005884	0.026561
215004	0.006255	0.032650
215005	0.007328	0.039739
215006	0.008627	0.048024
215007	0.010449	0.057971
215008	0.011167	0.068491
215009	0.007793	0.075750
215010	0.009372	0.084412
215011	0.009673	0.093268
215012	0.011568	0.103757
215013	0.012961	0.115373
215014	0.009753	0.124001
215015	0.008375	0.131337
215016	0.011478	0.141307

[165]:

```
pd_summary = (
    test_scored[test_scored["month_on_book"].isin([6, 12, 24, 36])]
    .groupby("month_on_book")
```

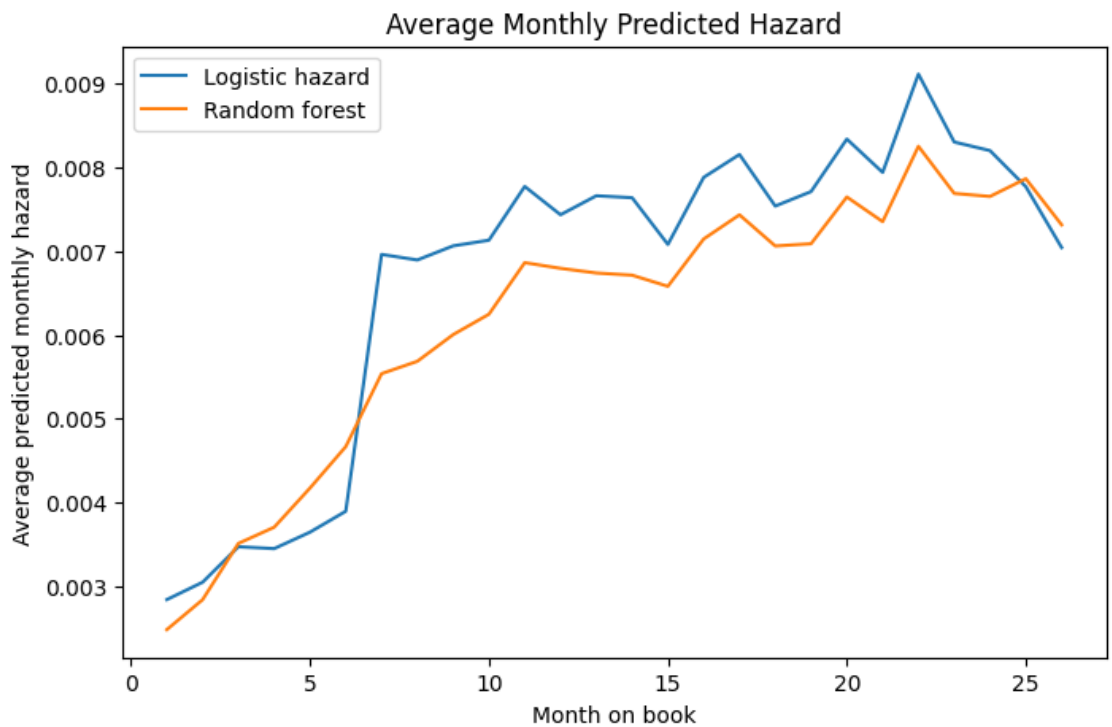
```
    .agg(
        avg_cum_pd_logit=("cum_pd_logit", "mean"),
        avg_cum_pd_rf=("cum_pd_rf", "mean"),
        loans=("id", "nunique"),
    )
    .reset_index()
)

pd_summary
```

[165]:

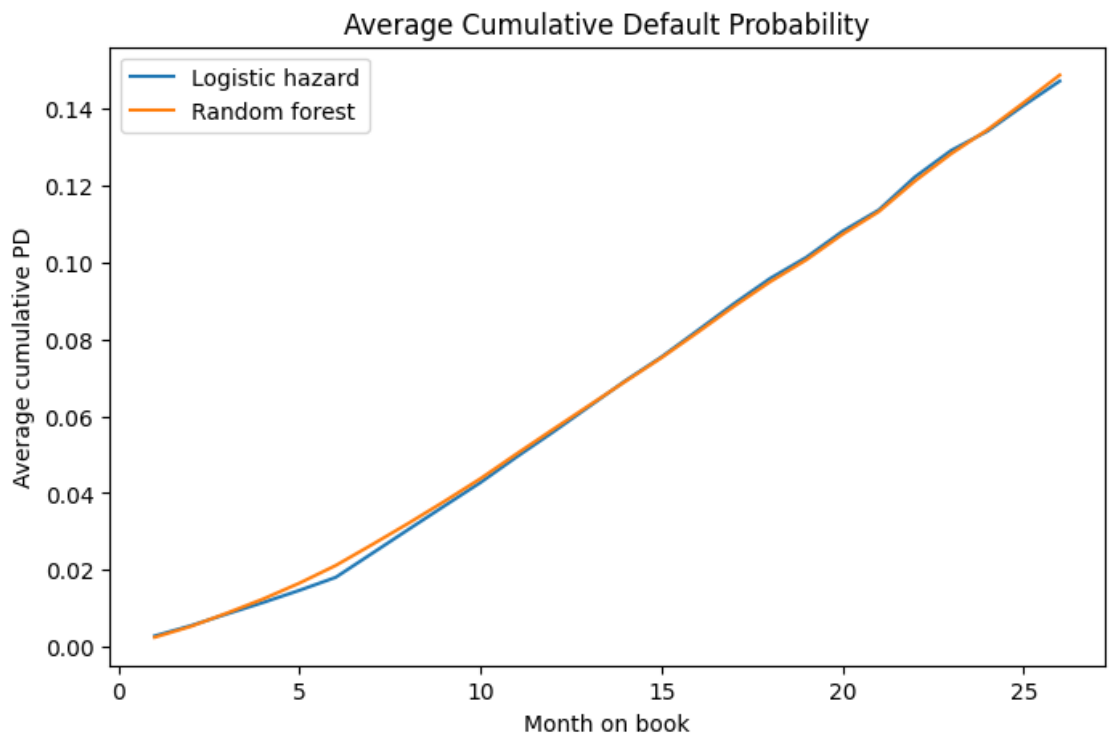
	month_on_book	avg_cum_pd_logit	avg_cum_pd_rf	loans
0	6	0.018068	0.021156	29591
1	12	0.055819	0.056530	17202
2	24	0.134077	0.134438	2001

[166]:

```
plot_hazard = (
    test_scored.groupby("month_on_book")[["pred_hazard_logit",␣
 ↪"pred_hazard_rf"]]
    .mean()
    .reset_index()
)

plt.figure(figsize=(8, 5))
plt.plot(plot_hazard["month_on_book"], plot_hazard["pred_hazard_logit"],␣
 ↪label="Logistic hazard")
plt.plot(plot_hazard["month_on_book"], plot_hazard["pred_hazard_rf"],␣
 ↪label="Random forest")
plt.xlabel("Month on book")
plt.ylabel("Average predicted monthly hazard")
plt.title("Average Monthly Predicted Hazard")
plt.legend()
plt.show()
```



[167]:
```
plot_cum_pd = (
    test_scored.groupby("month_on_book")[["cum_pd_logit", "cum_pd_rf"]]
    .mean()
    .reset_index()
)

plt.figure(figsize=(8, 5))
plt.plot(plot_cum_pd["month_on_book"], plot_cum_pd["cum_pd_logit"],␣
↪label="Logistic hazard")
plt.plot(plot_cum_pd["month_on_book"], plot_cum_pd["cum_pd_rf"], label="Random␣
↪forest")
plt.xlabel("Month on book")
plt.ylabel("Average cumulative PD")
plt.title("Average Cumulative Default Probability")
plt.legend()
plt.show()
```



[168]:
```
loan_12m = test_scored[test_scored["month_on_book"] == 12].copy()

print("Loans with 12-month horizon rows:", loan_12m["id"].nunique())
loan_12m[["id", "cum_pd_logit", "cum_pd_rf"]].head()
```

```
Loans with 12-month horizon rows: 17202
```

[168]:

	id	cum_pd_logit	cum_pd_rf
215008	100001365	0.070674	0.068491
539498	100001532	0.024204	0.039324
1473383	100001557	0.066775	0.071528
62778	100002096	0.060935	0.058785
1746963	100002651	0.096748	0.065673

[169]:
```
LGD = 0.60

loan_12m["ead_proxy"] = loan_12m["loan_amnt"]
loan_12m["expected_loss_12m_logit"] = loan_12m["cum_pd_logit"] * LGD *␣
↪loan_12m["ead_proxy"]
loan_12m["expected_loss_12m_rf"] = loan_12m["cum_pd_rf"] * LGD *␣
↪loan_12m["ead_proxy"]

loan_12m[[
```

```
    "id",
    "loan_amnt",
    "cum_pd_logit",
    "cum_pd_rf",
    "expected_loss_12m_logit",
    "expected_loss_12m_rf"
]].head()
```

[169]:

```
                id  loan_amnt  cum_pd_logit  cum_pd_rf  \
215008   100001365    20000.0      0.070674   0.068491
539498   100001532    30000.0      0.024204   0.039324
1473383  100001557    28000.0      0.066775   0.071528
62778    100002096    20000.0      0.060935   0.058785
1746963  100002651    10000.0      0.096748   0.065673

         expected_loss_12m_logit  expected_loss_12m_rf
215008                848.084448            821.894501
539498                435.663889            707.838201
1473383              1121.819319           1201.665842
62778                 731.223079            705.425546
1746963               580.488633            394.039128
```

[170]:

```
funding_cost = 0.04
profit_margin = 0.03

loan_12m["el_rate_12m_logit"] = loan_12m["expected_loss_12m_logit"] /␣
↪loan_12m["loan_amnt"]
loan_12m["el_rate_12m_rf"] = loan_12m["expected_loss_12m_rf"] /␣
↪loan_12m["loan_amnt"]

loan_12m["required_rate_logit"] = funding_cost + profit_margin +␣
↪loan_12m["el_rate_12m_logit"]
loan_12m["required_rate_rf"] = funding_cost + profit_margin +␣
↪loan_12m["el_rate_12m_rf"]

loan_12m[[
    "id",
    "required_rate_logit",
    "required_rate_rf"
]].head()
```

[170]:

```
                id  required_rate_logit  required_rate_rf
215008   100001365             0.112404          0.111095
539498   100001532             0.084522          0.093595
1473383  100001557             0.110065          0.112917
62778    100002096             0.106561          0.105271
1746963  100002651             0.128049          0.109404
```

[171]:
```
loan_12m["risk_band_logit"] = pd.qcut(
    loan_12m["cum_pd_logit"],
    q=5,
    labels=["Very Low", "Low", "Medium", "High", "Very High"]
)

reserve_summary_logit = (
    loan_12m.groupby("risk_band_logit")
    .agg(
        loans=("id", "count"),
        avg_pd_12m=("cum_pd_logit", "mean"),
        total_expected_loss=("expected_loss_12m_logit", "sum"),
        avg_required_rate=("required_rate_logit", "mean"),
    )
    .reset_index()
)

reserve_summary_logit
```

[171]:

	risk_band_logit	loans	avg_pd_12m	total_expected_loss	avg_required_rate
0	Very Low	3441	0.021078	6.062887e+05	0.082647
1	Low	3440	0.036170	1.043388e+06	0.091702
2	Medium	3440	0.048616	1.456756e+06	0.099170
3	High	3440	0.064421	2.025379e+06	0.108653
4	Very High	3441	0.108806	4.177691e+06	0.135283

[172]:
```
loan_12m["risk_band_rf"] = pd.qcut(
    loan_12m["cum_pd_rf"],
    q=5,
    labels=["Very Low", "Low", "Medium", "High", "Very High"]
)

reserve_summary_rf = (
    loan_12m.groupby("risk_band_rf")
    .agg(
        loans=("id", "count"),
        avg_pd_12m=("cum_pd_rf", "mean"),
        total_expected_loss=("expected_loss_12m_rf", "sum"),
        avg_required_rate=("required_rate_rf", "mean"),
    )
    .reset_index()
)

reserve_summary_rf
```

[172]:

	risk_band_rf	loans	avg_pd_12m	total_expected_loss	avg_required_rate
0	Very Low	3441	0.033973	9.751707e+05	0.090384

```
1          Low   3440    0.043509         1.288540e+06           0.096106
2       Medium   3440    0.052378         1.587341e+06           0.101427
3         High   3440    0.063665         2.000444e+06           0.108199
4    Very High   3441    0.089123         3.303155e+06           0.123474
```

[173]:
```
test_scored_path = PROCESSED_DIR / "test_panel_scored.csv"
loan_12m_path = PROCESSED_DIR / "loan_level_12m_outputs.csv"
results_path = PROCESSED_DIR / "model_comparison_results.csv"

test_scored.to_csv(test_scored_path, index=False)
loan_12m.to_csv(loan_12m_path, index=False)
results_df.to_csv(results_path, index=False)

print(f"Saved scored panel to: {test_scored_path}")
print(f"Saved 12m loan outputs to: {loan_12m_path}")
print(f"Saved model comparison to: {results_path}")
```

```
Saved scored panel to: data/processed/test_panel_scored.csv
Saved 12m loan outputs to: data/processed/loan_level_12m_outputs.csv
Saved model comparison to: data/processed/model_comparison_results.csv
```

[174]:
```
print("Model comparison:")
display(results_df)

print("\n12-month cumulative PD summary:")
display(pd_summary)

print("\nReserve summary (logistic hazard):")
display(reserve_summary_logit)
```

```
Model comparison:

             model  valid_roc_auc  valid_pr_auc  test_roc_auc  test_pr_auc
0  logistic_hazard       0.664451      0.013514      0.670544     0.008342
1    random_forest       0.656837      0.012907      0.665284     0.008312


12-month cumulative PD summary:

   month_on_book  avg_cum_pd_logit  avg_cum_pd_rf  loans
0              6          0.018068       0.021156  29591
1             12          0.055819       0.056530  17202
2             24          0.134077       0.134438   2001


Reserve summary (logistic hazard):

  risk_band_logit  loans  avg_pd_12m  total_expected_loss  avg_required_rate
0        Very Low   3441    0.021078         6.062887e+05           0.082647
1             Low   3440    0.036170         1.043388e+06           0.091702
2          Medium   3440    0.048616         1.456756e+06           0.099170
```

```
3            High   3440    0.064421         2.025379e+06           0.108653
4       Very High   3441    0.108806         4.177691e+06           0.135283
```

[175]:
```python
loan_12m["risk_band_rf"] = pd.qcut(
    loan_12m["cum_pd_rf"],
    q=5,
    labels=["Very Low", "Low", "Medium", "High", "Very High"]
)

reserve_summary_rf = (
    loan_12m.groupby("risk_band_rf")
    .agg(
        loans=("id", "count"),
        avg_pd_12m=("cum_pd_rf", "mean"),
        total_expected_loss=("expected_loss_12m_rf", "sum"),
        avg_required_rate=("required_rate_rf", "mean"),
    )
    .reset_index()
)

reserve_summary_rf
```

[175]:

	risk_band_rf	loans	avg_pd_12m	total_expected_loss	avg_required_rate
0	Very Low	3441	0.033973	9.751707e+05	0.090384
1	Low	3440	0.043509	1.288540e+06	0.096106
2	Medium	3440	0.052378	1.587341e+06	0.101427
3	High	3440	0.063665	2.000444e+06	0.108199
4	Very High	3441	0.089123	3.303155e+06	0.123474

14 Looking at test

[176]:
```python
test_loans = df[df["issue_d"] >= "2017-01-01"].copy()

loan_obs = test_loans[["id", "duration_months", "event"]].drop_duplicates().
 ↪copy()

print(loan_obs.shape)
loan_obs.head()
```

```
(37407, 3)
```

[176]:

	id	duration_months	event
2066630	125356772	10	0
477199	128490686	13	0
641990	109655666	14	0
507598	127049010	14	0
2129008	121746633	17	0

[177]:
```
max_mob = int(test_scored["month_on_book"].max())
months = np.arange(1, max_mob + 1)

n_loans = loan_obs["id"].nunique()

observed_curve = pd.DataFrame({"month_on_book": months})

observed_curve["observed_cum_pd"] = [
    ((loan_obs["event"] == 1) & (loan_obs["duration_months"] <= m)).sum() /␣
 ↪n_loans
    for m in months
]

observed_curve.head(15)
```

[177]:

	month_on_book	observed_cum_pd
0	1	0.001845
1	2	0.004251
2	3	0.007031
3	4	0.010934
4	5	0.014890
5	6	0.018820
6	7	0.022696
7	8	0.026439
8	9	0.029914
9	10	0.033336
10	11	0.036678
11	12	0.039832
12	13	0.041917
13	14	0.043976
14	15	0.045633

[178]:
```
plot_cum_pd = (
    test_scored.groupby("month_on_book")[["cum_pd_logit", "cum_pd_rf"]]
    .mean()
    .reset_index()
)
```

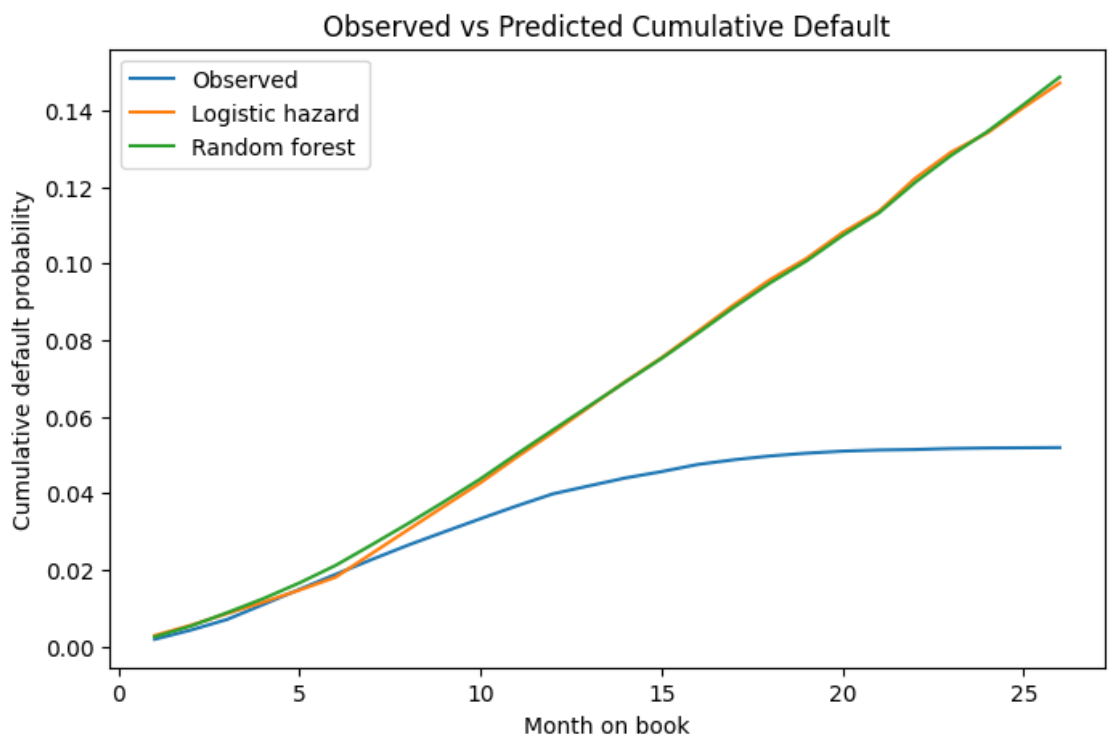
[179]:
```
compare_curve = observed_curve.merge(
    plot_cum_pd,
    on="month_on_book",
    how="inner"
)

compare_curve.head()
```

[179]:

	month_on_book	observed_cum_pd	cum_pd_logit	cum_pd_rf
0	1	0.001845	0.002840	0.002482
1	2	0.004251	0.005515	0.005312
2	3	0.007031	0.008556	0.008801
3	4	0.010934	0.011540	0.012466
4	5	0.014890	0.014696	0.016579

[180]:

```
plt.figure(figsize=(8, 5))
plt.plot(compare_curve["month_on_book"], compare_curve["observed_cum_pd"],␣
 ↪label="Observed")
plt.plot(compare_curve["month_on_book"], compare_curve["cum_pd_logit"],␣
 ↪label="Logistic hazard")
plt.plot(compare_curve["month_on_book"], compare_curve["cum_pd_rf"],␣
 ↪label="Random forest")
plt.xlabel("Month on book")
plt.ylabel("Cumulative default probability")
plt.title("Observed vs Predicted Cumulative Default")
plt.legend()
plt.show()
```



[181]:

```
milestones = [6, 12, 18, 24]

calibration_table = compare_curve[
```

```
    compare_curve["month_on_book"].isin(milestones)
].copy()

calibration_table["logit_error"] = (
    calibration_table["cum_pd_logit"] - calibration_table["observed_cum_pd"]
)

calibration_table["rf_error"] = (
    calibration_table["cum_pd_rf"] - calibration_table["observed_cum_pd"]
)

calibration_table
```

[181]:

```
    month_on_book  observed_cum_pd  cum_pd_logit  cum_pd_rf  logit_error  \
5               6         0.018820      0.018068   0.021156    -0.000752
11             12         0.039832      0.055819   0.056530     0.015987
17             18         0.049723      0.095799   0.094925     0.046075
23             24         0.051808      0.134077   0.134438     0.082268

    rf_error
5   0.002336
11  0.016698
17  0.045201
23  0.082629
```

[182]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from lifelines import KaplanMeierFitter

# ==========================================================
# 1. Build loan-level TEST set for Kaplan-Meier
# ==========================================================

# Make sure issue_d is datetime
df["issue_d"] = pd.to_datetime(df["issue_d"], errors="coerce")

# Match your test split
test_loans = df[df["issue_d"] >= "2017-01-01"].copy()

# Keep one row per loan with survival inputs
loan_obs = (
    test_loans[["id", "duration_months", "event"]]
    .drop_duplicates(subset=["id"])
    .copy()
)
```

```
# Clean types
loan_obs["duration_months"] = pd.to_numeric(loan_obs["duration_months"],␣
↪errors="coerce")
loan_obs["event"] = pd.to_numeric(loan_obs["event"], errors="coerce")

# Drop bad rows if any
loan_obs = loan_obs.dropna(subset=["id", "duration_months", "event"]).copy()

# Optional: force valid ranges/types
loan_obs["duration_months"] = loan_obs["duration_months"].astype(int)
loan_obs["event"] = loan_obs["event"].astype(int)

print("Loan-level test sample shape:", loan_obs.shape)
print(loan_obs.head())


# ==========================================================
# 2. Fit Kaplan-Meier on observed test-loan outcomes
# ==========================================================

kmf = KaplanMeierFitter()

kmf.fit(
    durations=loan_obs["duration_months"],
    event_observed=loan_obs["event"],
    label="Kaplan-Meier survival"
)

# Convert survival curve to cumulative default curve
km_curve = kmf.survival_function_.reset_index()
km_curve.columns = ["month_on_book", "survival_prob"]
km_curve["observed_cum_pd_km"] = 1 - km_curve["survival_prob"]

print("\nKaplan-Meier curve:")
print(km_curve.head(15))


# ==========================================================
# 3. Build average model-predicted cumulative PD curve
# ==========================================================

plot_cum_pd = (
    test_scored.groupby("month_on_book")[["cum_pd_logit", "cum_pd_rf"]]
    .mean()
    .reset_index()
    .sort_values("month_on_book")
)
```

```python
print("\nPredicted cumulative PD curve:")
print(plot_cum_pd.head(15))


# ==========================================================
# 4. Merge KM observed curve with model curves
# ==========================================================

compare_curve = pd.merge(
    plot_cum_pd,
    km_curve[["month_on_book", "observed_cum_pd_km"]],
    on="month_on_book",
    how="left"
).sort_values("month_on_book")

print("\nMerged comparison curve:")
print(compare_curve.head(15))


# ==========================================================
# 5. Plot observed KM cumulative default vs predicted curves
# ==========================================================

plt.figure(figsize=(10, 6))
plt.plot(
    compare_curve["month_on_book"],
    compare_curve["observed_cum_pd_km"],
    label="Observed cumulative PD (Kaplan-Meier)"
)
plt.plot(
    compare_curve["month_on_book"],
    compare_curve["cum_pd_logit"],
    label="Predicted cumulative PD (Logistic hazard)"
)
plt.plot(
    compare_curve["month_on_book"],
    compare_curve["cum_pd_rf"],
    label="Predicted cumulative PD (Random forest)"
)

plt.xlabel("Month on book")
plt.ylabel("Cumulative default probability")
plt.title("Observed vs Predicted Cumulative Default Probability")
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```

```python
# ==========================================================
# 6. Optional: milestone calibration table
# ==========================================================

milestones = [6, 12, 18, 24]

calibration_table = (
    compare_curve[compare_curve["month_on_book"].isin(milestones)]
    .copy()
)

calibration_table["logit_error"] = (
    calibration_table["cum_pd_logit"] - calibration_table["observed_cum_pd_km"]
)

calibration_table["rf_error"] = (
    calibration_table["cum_pd_rf"] - calibration_table["observed_cum_pd_km"]
)

print("\nCalibration table:")
print(calibration_table[[
    "month_on_book",
    "observed_cum_pd_km",
    "cum_pd_logit",
    "cum_pd_rf",
    "logit_error",
    "rf_error"
]])
```

```
Loan-level test sample shape: (37407, 3)
                id  duration_months  event
2066630  125356772               10      0
477199   128490686               13      0
641990   109655666               14      0
507598   127049010               14      0
2129008  121746633               17      0

Kaplan-Meier curve:
    month_on_book  survival_prob  observed_cum_pd_km
0             0.0       1.000000            0.000000
1             1.0       0.998155            0.001845
2             2.0       0.995704            0.004296
3             3.0       0.992835            0.007165
4             4.0       0.988582            0.011418
5             5.0       0.983980            0.016020
6             6.0       0.979092            0.020908
```

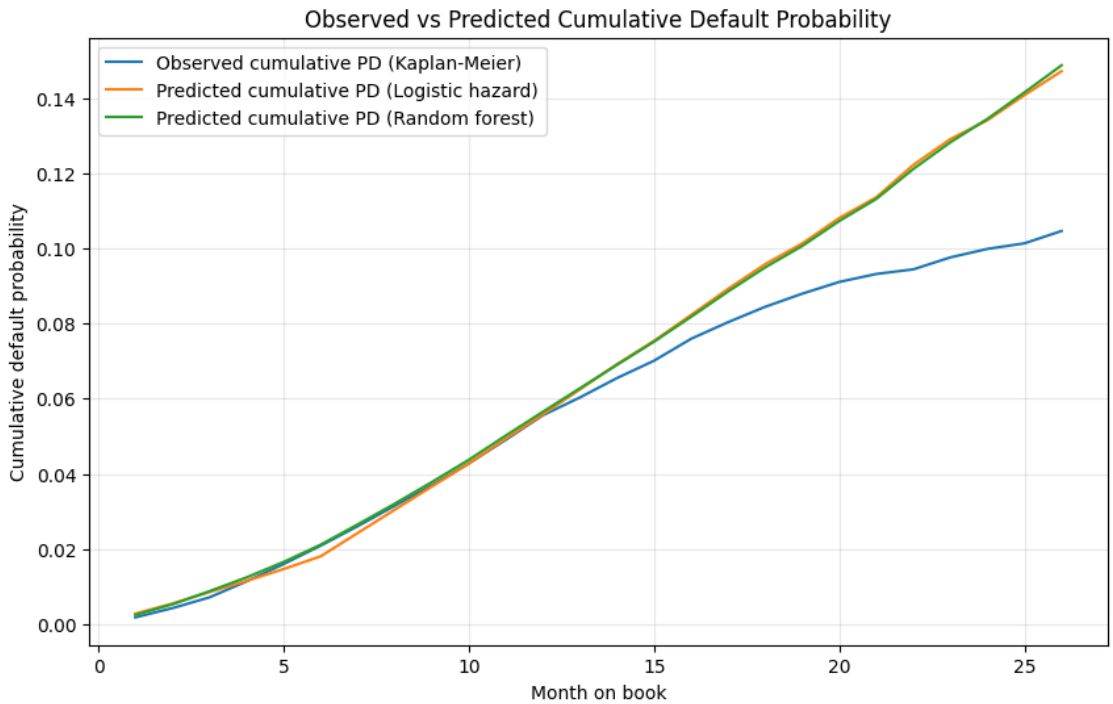
7	7.0	0.973916	0.026084
8	8.0	0.968488	0.031512
9	9.0	0.963048	0.036952
10	10.0	0.957203	0.042797
11	11.0	0.950921	0.049079
12	12.0	0.944398	0.055602
13	13.0	0.939666	0.060334
14	14.0	0.934506	0.065494

Predicted cumulative PD curve:

	month_on_book	cum_pd_logit	cum_pd_rf
0	1	0.002840	0.002482
1	2	0.005515	0.005312
2	3	0.008556	0.008801
3	4	0.011540	0.012466
4	5	0.014696	0.016579
5	6	0.018068	0.021156
6	7	0.024264	0.026554
7	8	0.030465	0.032077
8	9	0.036624	0.037812
9	10	0.042731	0.043751
10	11	0.049393	0.050212
11	12	0.055819	0.056530
12	13	0.062478	0.062783
13	14	0.069183	0.069004
14	15	0.075383	0.075176

Merged comparison curve:

	month_on_book	cum_pd_logit	cum_pd_rf	observed_cum_pd_km
0	1	0.002840	0.002482	0.001845
1	2	0.005515	0.005312	0.004296
2	3	0.008556	0.008801	0.007165
3	4	0.011540	0.012466	0.011418
4	5	0.014696	0.016579	0.016020
5	6	0.018068	0.021156	0.020908
6	7	0.024264	0.026554	0.026084
7	8	0.030465	0.032077	0.031512
8	9	0.036624	0.037812	0.036952
9	10	0.042731	0.043751	0.042797
10	11	0.049393	0.050212	0.049079
11	12	0.055819	0.056530	0.055602
12	13	0.062478	0.062783	0.060334
13	14	0.069183	0.069004	0.065494
14	15	0.075383	0.075176	0.070102




```
Calibration table:
    month_on_book  observed_cum_pd_km  cum_pd_logit  cum_pd_rf  logit_error  \
5               6            0.020908      0.018068   0.021156    -0.002840
11             12            0.055602      0.055819   0.056530     0.000217
17             18            0.084446      0.095799   0.094925     0.011353
23             24            0.099847      0.134077   0.134438     0.034230

    rf_error
5   0.000247
11  0.000928
17  0.010479
23  0.034591
```

15 Proportional Hazard??

[183]:
```
# ========================
# Cox PH model: clean, loan-level setup
# ========================

import numpy as np
import pandas as pd
from lifelines import CoxPHFitter
from lifelines.utils import concordance_index
```

```python
cox_num_features = [
    "loan_amnt",
    "term_num",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "revol_util",
    "fico_avg",
]

cox_cat_features = [
    "verification_status",
    "purpose",
    "home_ownership",
]

required_cols = ["duration_months", "event", "issue_d"] + cox_num_features +␣
 ↪cox_cat_features

missing_cols = [c for c in required_cols if c not in df.columns]
if missing_cols:
    raise ValueError(f"Missing required columns in df: {missing_cols}")

loan_df = df[required_cols].copy()

# 1) Basic cleaning
loan_df["issue_d"] = pd.to_datetime(loan_df["issue_d"], errors="coerce")
loan_df["duration_months"] = pd.to_numeric(loan_df["duration_months"],␣
 ↪errors="coerce")
loan_df["event"] = pd.to_numeric(loan_df["event"], errors="coerce")

for col in cox_num_features:
    loan_df[col] = pd.to_numeric(loan_df[col], errors="coerce")

loan_df = loan_df.dropna(subset=["issue_d", "duration_months", "event"]).copy()
loan_df = loan_df[loan_df["duration_months"] > 0].copy()

# 2) Train / valid / test split at loan level
train_loans = loan_df[loan_df["issue_d"] < "2016-01-01"].copy()
valid_loans = loan_df[(loan_df["issue_d"] >= "2016-01-01") &␣
 ↪(loan_df["issue_d"] < "2017-01-01")].copy()
test_loans  = loan_df[loan_df["issue_d"] >= "2017-01-01"].copy()

print("Loan-level shapes before dummy encoding:")
print("Train:", train_loans.shape)
```

```python
print("Valid:", valid_loans.shape)
print("Test :", test_loans.shape)

# 3) One-hot encode categoricals with drop_first=True
def make_cox_matrix(x, cat_cols):
    out = pd.get_dummies(x, columns=cat_cols, drop_first=True)
    bool_cols = out.select_dtypes(include=["bool"]).columns
    if len(bool_cols) > 0:
        out[bool_cols] = out[bool_cols].astype(int)
    return out

train_cox = make_cox_matrix(train_loans.drop(columns=["issue_d"]),␣
 ↪cox_cat_features)
valid_cox = make_cox_matrix(valid_loans.drop(columns=["issue_d"]),␣
 ↪cox_cat_features)
test_cox  = make_cox_matrix(test_loans.drop(columns=["issue_d"]),␣
 ↪cox_cat_features)

# 4) Align columns across splits
feature_cols_train = [c for c in train_cox.columns if c not in␣
 ↪["duration_months", "event"]]

valid_cox = valid_cox.reindex(columns=["duration_months", "event"] +␣
 ↪feature_cols_train, fill_value=0)
test_cox  = test_cox.reindex(columns=["duration_months", "event"] +␣
 ↪feature_cols_train, fill_value=0)

# 5) Remove constant columns from training data
constant_cols = [c for c in feature_cols_train if train_cox[c].
 ↪nunique(dropna=False) <= 1]
if constant_cols:
    train_cox = train_cox.drop(columns=constant_cols)
    valid_cox = valid_cox.drop(columns=constant_cols, errors="ignore")
    test_cox  = test_cox.drop(columns=constant_cols, errors="ignore")

feature_cols = [c for c in train_cox.columns if c not in ["duration_months",␣
 ↪"event"]]

# 6) Check NaNs and drop incomplete rows
nan_counts = train_cox[["duration_months", "event"] + feature_cols].isna().sum()
nan_counts = nan_counts[nan_counts > 0]

if len(nan_counts) > 0:
    print("\nColumns with NaNs in training data:")
    print(nan_counts.sort_values(ascending=False))
```

```
train_cox = train_cox.dropna(subset=["duration_months", "event"] +␣
 ↪feature_cols).copy()
valid_cox = valid_cox.dropna(subset=["duration_months", "event"] +␣
 ↪feature_cols).copy()
test_cox  = test_cox.dropna(subset=["duration_months", "event"] + feature_cols).
 ↪copy()

print("\nLoan-level shapes after dropping NaNs:")
print("Train:", train_cox.shape)
print("Valid:", valid_cox.shape)
print("Test :", test_cox.shape)

# 7) Fit Cox PH with penalization
cph = CoxPHFitter(penalizer=0.1)
cph.fit(
    train_cox[["duration_months", "event"] + feature_cols],
    duration_col="duration_months",
    event_col="event",
    show_progress=False,
)

print("\nCox model fit complete.")
print("Number of features used:", len(feature_cols))

# 8) Evaluate with concordance index
train_partial_hazard = cph.predict_partial_hazard(train_cox[feature_cols]).
 ↪values.ravel()
valid_partial_hazard = cph.predict_partial_hazard(valid_cox[feature_cols]).
 ↪values.ravel()
test_partial_hazard  = cph.predict_partial_hazard(test_cox[feature_cols]).
 ↪values.ravel()

train_cindex = concordance_index(
    train_cox["duration_months"], -train_partial_hazard, train_cox["event"]
)
valid_cindex = concordance_index(
    valid_cox["duration_months"], -valid_partial_hazard, valid_cox["event"]
)
test_cindex = concordance_index(
    test_cox["duration_months"], -test_partial_hazard, test_cox["event"]
)

print("\nConcordance indices:")
print(f"Train C-index: {train_cindex:.4f}")
print(f"Valid C-index: {valid_cindex:.4f}")
print(f"Test  C-index: {test_cindex:.4f}")
```

```python
# 9) Show coefficient summary
summary_cols = ["coef", "exp(coef)", "se(coef)", "p"]
available_summary_cols = [c for c in summary_cols if c in cph.summary.columns]

print("\nTop positive-risk features:")
display(cph.summary[available_summary_cols].sort_values("coef",␣
↪ascending=False).head(15))

print("\nTop negative-risk features:")
display(cph.summary[available_summary_cols].sort_values("coef", ascending=True).
↪head(15))
```

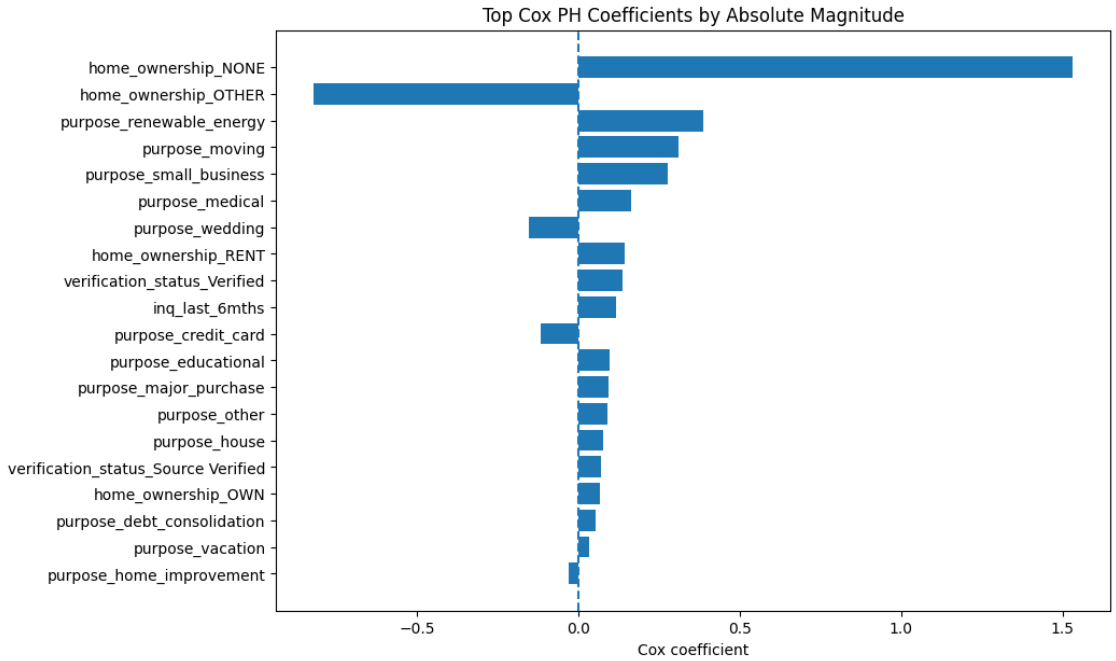
```
Loan-level shapes before dummy encoding:
Train: (35307, 14)
Valid: (17282, 14)
Test : (37407, 14)

Columns with NaNs in training data:
revol_util    15
dti            1
dtype: int64

Loan-level shapes after dropping NaNs:
Train: (35291, 29)
Valid: (17271, 29)
Test : (37305, 29)

Cox model fit complete.
Number of features used: 27

Concordance indices:
Train C-index: 0.6398
Valid C-index: 0.6369
Test  C-index: 0.6563

Top positive-risk features:
                                        coef  exp(coef)  se(coef)  \
covariate
home_ownership_NONE                 1.530817   4.621953  0.605168
purpose_renewable_energy            0.387797   1.473731  0.445817
purpose_moving                      0.311358   1.365278  0.130846
purpose_small_business              0.276367   1.318332  0.089690
purpose_medical                     0.163190   1.177261  0.099770
home_ownership_RENT                 0.143577   1.154396  0.021458
verification_status_Verified        0.136554   1.146317  0.023579
inq_last_6mths                      0.116474   1.123528  0.009849
```

```
purpose_educational                  0.096384   1.101182  0.596629
purpose_major_purchase               0.092936   1.097391  0.076038
purpose_other                        0.089689   1.093834  0.050956
purpose_house                        0.078069   1.081198  0.154310
verification_status_Source Verified  0.069375   1.071838  0.022884
home_ownership_OWN                   0.067671   1.070013  0.034873
purpose_debt_consolidation           0.054032   1.055518  0.025649

                                                   p
covariate
home_ownership_NONE                  1.142005e-02
purpose_renewable_energy             3.843779e-01
purpose_moving                       1.733260e-02
purpose_small_business               2.060507e-03
purpose_medical                      1.019091e-01
home_ownership_RENT                  2.215458e-11
verification_status_Verified         6.981845e-09
inq_last_6mths                       2.857451e-32
purpose_educational                  8.716620e-01
purpose_major_purchase               2.216211e-01
purpose_other                        7.838525e-02
purpose_house                        6.129103e-01
verification_status_Source Verified  2.432260e-03
home_ownership_OWN                   5.232224e-02
purpose_debt_consolidation           3.515667e-02


Top negative-risk features:
                                         coef  exp(coef)      se(coef)  \
covariate
home_ownership_OTHER                -0.819261   0.440757  8.863075e-01
purpose_wedding                     -0.154392   0.856936  2.266581e-01
purpose_credit_card                 -0.115797   0.890656  2.864385e-02
purpose_home_improvement            -0.029170   0.971251  4.869403e-02
fico_avg                            -0.005582   0.994434  3.792156e-04
delinq_2yrs                         -0.000519   0.999481  1.179691e-02
revol_util                          -0.000263   0.999737  4.587116e-04
annual_inc                          -0.000001   0.999999  2.266994e-07
loan_amnt                            0.000005   1.000005  1.336761e-06
term_num                             0.009274   1.009317  9.508547e-04
dti                                  0.010765   1.010824  1.243568e-03
purpose_vacation                     0.033559   1.034129  1.528523e-01
purpose_debt_consolidation           0.054032   1.055518  2.564936e-02
home_ownership_OWN                   0.067671   1.070013  3.487329e-02
verification_status_Source Verified  0.069375   1.071838  2.288365e-02

                                                   p
covariate
```

```
home_ownership_OTHER                 3.553028e-01
purpose_wedding                      4.957654e-01
purpose_credit_card                  5.285244e-05
purpose_home_improvement             5.491404e-01
fico_avg                             4.853111e-49
delinq_2yrs                          9.648758e-01
revol_util                           5.663093e-01
annual_inc                           1.520564e-10
loan_amnt                            5.659258e-04
term_num                             1.794056e-22
dti                                  4.847627e-18
purpose_vacation                     8.262177e-01
purpose_debt_consolidation           3.515667e-02
home_ownership_OWN                   5.232224e-02
verification_status_Source Verified  2.432260e-03
```

[184]:

```
# ========================
# Cox coefficient charts
# ========================

cox_summary = cph.summary.copy()

# keep only the columns we care about
cols_keep = [c for c in ["coef", "exp(coef)", "se(coef)", "p"] if c in␣
 ↪cox_summary.columns]
cox_summary = cox_summary[cols_keep].copy()
cox_summary["feature"] = cox_summary.index

# Sort by absolute effect size
cox_summary["abs_coef"] = cox_summary["coef"].abs()
cox_plot = cox_summary.sort_values("abs_coef", ascending=False).head(20).copy()

plt.figure(figsize=(10, 7))
plt.barh(cox_plot["feature"][::-1], cox_plot["coef"][::-1])
plt.axvline(0, linestyle="--")
plt.xlabel("Cox coefficient")
plt.title("Top Cox PH Coefficients by Absolute Magnitude")
plt.show()

display(cox_summary.sort_values("p").head(20))
```




```
                                     coef  exp(coef)      se(coef)  \
covariate
fico_avg                          -0.005582   0.994434  3.792156e-04
inq_last_6mths                     0.116474   1.123528  9.848766e-03
term_num                           0.009274   1.009317  9.508547e-04
dti                                0.010765   1.010824  1.243568e-03
home_ownership_RENT                0.143577   1.154396  2.145803e-02
annual_inc                        -0.000001   0.999999  2.266994e-07
verification_status_Verified       0.136554   1.146317  2.357899e-02
purpose_credit_card               -0.115797   0.890656  2.864385e-02
loan_amnt                          0.000005   1.000005  1.336761e-06
purpose_small_business             0.276367   1.318332  8.968972e-02
verification_status_Source Verified  0.069375   1.071838  2.288365e-02
home_ownership_NONE                1.530817   4.621953  6.051676e-01
purpose_moving                     0.311358   1.365278  1.308462e-01
purpose_debt_consolidation         0.054032   1.055518  2.564936e-02
home_ownership_OWN                 0.067671   1.070013  3.487329e-02
purpose_other                      0.089689   1.093834  5.095570e-02
purpose_medical                    0.163190   1.177261  9.976978e-02
purpose_major_purchase             0.092936   1.097391  7.603806e-02
home_ownership_OTHER              -0.819261   0.440757  8.863075e-01
purpose_renewable_energy           0.387797   1.473731  4.458169e-01

                                           p  \
covariate
fico_avg                        4.853111e-49
```

```
inq_last_6mths                      2.857451e-32
term_num                            1.794056e-22
dti                                 4.847627e-18
home_ownership_RENT                 2.215458e-11
annual_inc                          1.520564e-10
verification_status_Verified        6.981845e-09
purpose_credit_card                 5.285244e-05
loan_amnt                           5.659258e-04
purpose_small_business              2.060507e-03
verification_status_Source Verified 2.432260e-03
home_ownership_NONE                 1.142005e-02
purpose_moving                      1.733260e-02
purpose_debt_consolidation          3.515667e-02
home_ownership_OWN                  5.232224e-02
purpose_other                       7.838525e-02
purpose_medical                     1.019091e-01
purpose_major_purchase              2.216211e-01
home_ownership_OTHER                3.553028e-01
purpose_renewable_energy            3.843779e-01

                                                                  feature  \
covariate
fico_avg                                                        fico_avg
inq_last_6mths                                            inq_last_6mths
term_num                                                        term_num
dti                                                                  dti
home_ownership_RENT                                   home_ownership_RENT
annual_inc                                                     annual_inc
verification_status_Verified                verification_status_Verified
purpose_credit_card                                   purpose_credit_card
loan_amnt                                                       loan_amnt
purpose_small_business                              purpose_small_business
verification_status_Source Verified  verification_status_Source Verified
home_ownership_NONE                                   home_ownership_NONE
purpose_moving                                             purpose_moving
purpose_debt_consolidation                    purpose_debt_consolidation
home_ownership_OWN                                     home_ownership_OWN
purpose_other                                               purpose_other
purpose_medical                                           purpose_medical
purpose_major_purchase                              purpose_major_purchase
home_ownership_OTHER                                  home_ownership_OTHER
purpose_renewable_energy                            purpose_renewable_energy

                                    abs_coef
covariate
fico_avg                            0.005582
inq_last_6mths                      0.116474
term_num                            0.009274
```

```
dti                                   0.010765
home_ownership_RENT                   0.143577
annual_inc                            0.000001
verification_status_Verified          0.136554
purpose_credit_card                   0.115797
loan_amnt                             0.000005
purpose_small_business                0.276367
verification_status_Source Verified   0.069375
home_ownership_NONE                   1.530817
purpose_moving                        0.311358
purpose_debt_consolidation            0.054032
home_ownership_OWN                    0.067671
purpose_other                         0.089689
purpose_medical                       0.163190
purpose_major_purchase                0.092936
home_ownership_OTHER                  0.819261
purpose_renewable_energy              0.387797
```

[185]:

```python
# ========================
# Compare models
# ========================

comparison_rows = []

# Earlier results from your notebook if they still exist:
# hazard_results
# rf_results

if "hazard_results" in globals():
    comparison_rows.append({
        "model": "logistic_hazard_old",
        "valid_roc_auc": hazard_results.get("valid_roc_auc"),
        "valid_pr_auc": hazard_results.get("valid_pr_auc"),
        "test_roc_auc": hazard_results.get("test_roc_auc"),
        "test_pr_auc": hazard_results.get("test_pr_auc"),
        "valid_c_index": np.nan,
        "test_c_index": np.nan,
    })

if "rf_results" in globals():
    comparison_rows.append({
        "model": "random_forest_old",
        "valid_roc_auc": rf_results.get("valid_roc_auc"),
        "valid_pr_auc": rf_results.get("valid_pr_auc"),
        "test_roc_auc": rf_results.get("test_roc_auc"),
        "test_pr_auc": rf_results.get("test_pr_auc"),
        "valid_c_index": np.nan,
```

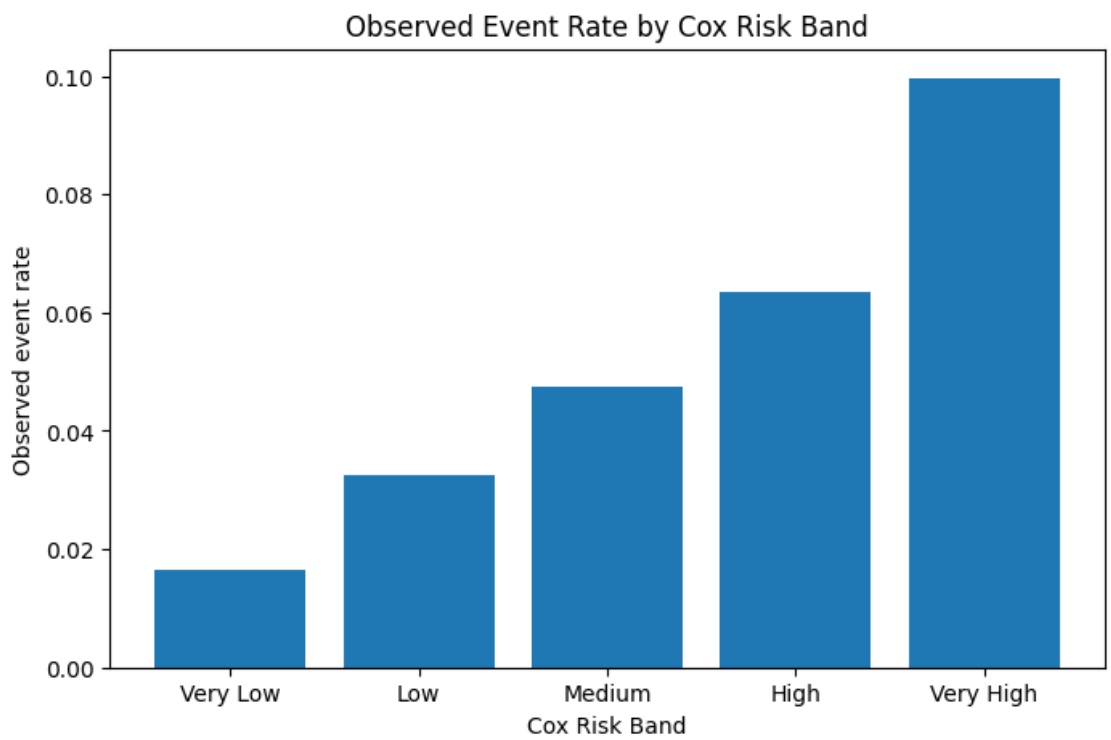
```
        "test_c_index": np.nan,
    })

comparison_rows.append({
    "model": "cox_ph_new",
    "valid_roc_auc": np.nan,
    "valid_pr_auc": np.nan,
    "test_roc_auc": np.nan,
    "test_pr_auc": np.nan,
    "valid_c_index": valid_cindex,
    "test_c_index": test_cindex,
})

model_compare_df = pd.DataFrame(comparison_rows)
display(model_compare_df)
```

```
                 model  valid_roc_auc  valid_pr_auc  test_roc_auc  \
0  logistic_hazard_old       0.664451      0.013514      0.670544
1    random_forest_old       0.656837      0.012907      0.665284
2           cox_ph_new            NaN           NaN           NaN

   test_pr_auc  valid_c_index  test_c_index
0     0.008342            NaN           NaN
1     0.008312            NaN           NaN
2          NaN       0.636881       0.65629
```

[186]:
```
# ========================
# Cox risk bands on test set
# ========================

cox_test_scored = test_loans.copy()
cox_test_scored = cox_test_scored.loc[test_cox.index].copy()
cox_test_scored["cox_partial_hazard"] = test_partial_hazard

cox_test_scored["cox_risk_band"] = pd.qcut(
    cox_test_scored["cox_partial_hazard"],
    q=5,
    labels=["Very Low", "Low", "Medium", "High", "Very High"]
)

cox_band_summary = (
    cox_test_scored.groupby("cox_risk_band")
    .agg(
        loans=("event", "count"),
        event_rate=("event", "mean"),
        avg_duration=("duration_months", "mean"),
        avg_partial_hazard=("cox_partial_hazard", "mean"),
```

```
    )
    .reset_index()
)

display(cox_band_summary)

plt.figure(figsize=(8, 5))
plt.bar(cox_band_summary["cox_risk_band"], cox_band_summary["event_rate"])
plt.xlabel("Cox Risk Band")
plt.ylabel("Observed event rate")
plt.title("Observed Event Rate by Cox Risk Band")
plt.show()
```

	cox_risk_band	loans	event_rate	avg_duration	avg_partial_hazard
0	Very Low	7461	0.016486	11.269267	0.580797
1	Low	7461	0.032435	11.470580	0.792375
2	Medium	7461	0.047581	11.778850	0.938601
3	High	7461	0.063530	12.080954	1.102321
4	Very High	7461	0.099585	11.945316	28.329924



[187]:
```
# ========================
# Cox cumulative PD over time vs prior models
# ========================
```

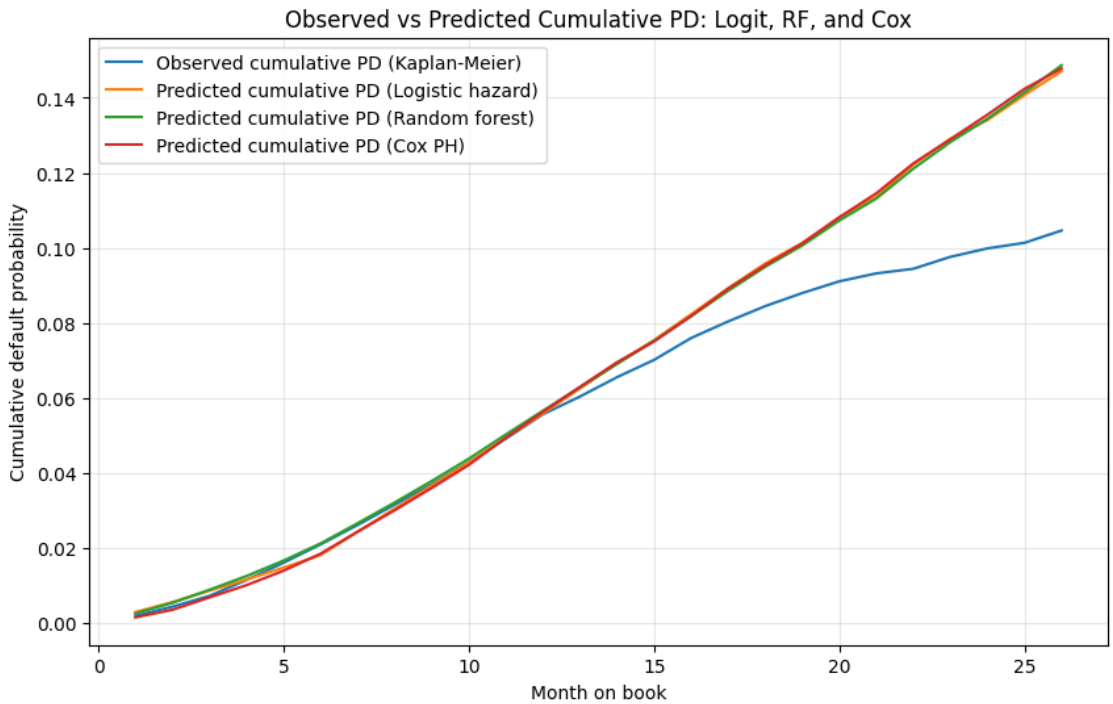
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ---------------------------------------------------
# 1) Build Cox predicted survival curves on test set
# ---------------------------------------------------
# test_cox: encoded loan-level test matrix used in Cox
# feature_cols: Cox feature columns
# cph: fitted CoxPHFitter
# compare_curve: earlier dataframe with month_on_book, observed_cum_pd_km,␣
↪cum_pd_logit, cum_pd_rf
# If compare_curve does not exist, this cell rebuilds it from your earlier␣
↪objects.

# Predict survival functions for each test loan
cox_surv = cph.predict_survival_function(test_cox[feature_cols])

# cox_surv rows are times, columns are loans
# Convert to cumulative PD
cox_cum_pd = 1 - cox_surv

# Average across loans at each time
cox_curve = cox_cum_pd.mean(axis=1).reset_index()
cox_curve.columns = ["month_on_book", "cum_pd_cox"]

# lifelines may return float times like 1.0, 2.0, ...
cox_curve["month_on_book"] = cox_curve["month_on_book"].astype(float)

# ---------------------------------------------------
# 2) Rebuild earlier comparison curve if needed
# ---------------------------------------------------
if "compare_curve" not in globals():
    # observed KM cumulative PD
    from lifelines import KaplanMeierFitter

    loan_obs = (
        df.loc[df["issue_d"] >= "2017-01-01", ["id", "duration_months",␣
↪"event"]]
        .drop_duplicates(subset=["id"])
        .copy()
    )
    loan_obs["duration_months"] = pd.to_numeric(loan_obs["duration_months"],␣
↪errors="coerce")
    loan_obs["event"] = pd.to_numeric(loan_obs["event"], errors="coerce")
    loan_obs = loan_obs.dropna(subset=["duration_months", "event"]).copy()
```

```
    loan_obs["duration_months"] = loan_obs["duration_months"].astype(int)
    loan_obs["event"] = loan_obs["event"].astype(int)

    kmf = KaplanMeierFitter()
    kmf.fit(
        durations=loan_obs["duration_months"],
        event_observed=loan_obs["event"],
        label="Kaplan-Meier survival"
    )

    km_curve = kmf.survival_function_.reset_index()
    km_curve.columns = ["month_on_book", "survival_prob"]
    km_curve["observed_cum_pd_km"] = 1 - km_curve["survival_prob"]

    # earlier logit/RF cumulative PD from panel test scores
    plot_cum_pd = (
        test_scored.groupby("month_on_book")[["cum_pd_logit", "cum_pd_rf"]]
        .mean()
        .reset_index()
        .sort_values("month_on_book")
    )

    compare_curve = pd.merge(
        plot_cum_pd,
        km_curve[["month_on_book", "observed_cum_pd_km"]],
        on="month_on_book",
        how="left"
    ).sort_values("month_on_book")

# ---------------------------------------------------
# 3) Merge Cox curve with prior curves
# ---------------------------------------------------
compare_curve_with_cox = pd.merge(
    compare_curve,
    cox_curve,
    on="month_on_book",
    how="left"
).sort_values("month_on_book")

display(compare_curve_with_cox.head(15))

# ---------------------------------------------------
# 4) Plot all cumulative PD curves together
# ---------------------------------------------------
plt.figure(figsize=(10, 6))
plt.plot(
    compare_curve_with_cox["month_on_book"],
```

```
    compare_curve_with_cox["observed_cum_pd_km"],
    label="Observed cumulative PD (Kaplan-Meier)"
)
plt.plot(
    compare_curve_with_cox["month_on_book"],
    compare_curve_with_cox["cum_pd_logit"],
    label="Predicted cumulative PD (Logistic hazard)"
)
plt.plot(
    compare_curve_with_cox["month_on_book"],
    compare_curve_with_cox["cum_pd_rf"],
    label="Predicted cumulative PD (Random forest)"
)
plt.plot(
    compare_curve_with_cox["month_on_book"],
    compare_curve_with_cox["cum_pd_cox"],
    label="Predicted cumulative PD (Cox PH)"
)

plt.xlabel("Month on book")
plt.ylabel("Cumulative default probability")
plt.title("Observed vs Predicted Cumulative PD: Logit, RF, and Cox")
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

# ---------------------------------------------------
# 5) Milestone calibration table including Cox
# ---------------------------------------------------
milestones = [6, 12, 18, 24]

calibration_table_all = (
    compare_curve_with_cox[compare_curve_with_cox["month_on_book"].
 ↪isin(milestones)]
    .copy()
)

calibration_table_all["logit_error"] = (
    calibration_table_all["cum_pd_logit"] -␣
 ↪calibration_table_all["observed_cum_pd_km"]
)
calibration_table_all["rf_error"] = (
    calibration_table_all["cum_pd_rf"] -␣
 ↪calibration_table_all["observed_cum_pd_km"]
)
calibration_table_all["cox_error"] = (
```

```
    calibration_table_all["cum_pd_cox"] -␣
 ↪calibration_table_all["observed_cum_pd_km"]
)

display(
    calibration_table_all[
        [
            "month_on_book",
            "observed_cum_pd_km",
            "cum_pd_logit",
            "cum_pd_rf",
            "cum_pd_cox",
            "logit_error",
            "rf_error",
            "cox_error",
        ]
    ]
)
```

	month_on_book	cum_pd_logit	cum_pd_rf	observed_cum_pd_km	cum_pd_cox
0	1	0.002840	0.002482	0.001845	0.001443
1	2	0.005515	0.005312	0.004296	0.003462
2	3	0.008556	0.008801	0.007165	0.006765
3	4	0.011540	0.012466	0.011418	0.010028
4	5	0.014696	0.016579	0.016020	0.013889
5	6	0.018068	0.021156	0.020908	0.018417
6	7	0.024264	0.026554	0.026084	0.024304
7	8	0.030465	0.032077	0.031512	0.030007
8	9	0.036624	0.037812	0.036952	0.035965
9	10	0.042731	0.043751	0.042797	0.042097
10	11	0.049393	0.050212	0.049079	0.049347
11	12	0.055819	0.056530	0.055602	0.056270
12	13	0.062478	0.062783	0.060334	0.062910
13	14	0.069183	0.069004	0.065494	0.069463
14	15	0.075383	0.075176	0.070102	0.075019




```
    month_on_book  observed_cum_pd_km  cum_pd_logit  cum_pd_rf  cum_pd_cox  \
5               6            0.020908      0.018068   0.021156    0.018417
11             12            0.055602      0.055819   0.056530    0.056270
17             18            0.084446      0.095799   0.094925    0.095351
23             24            0.099847      0.134077   0.134438    0.135520

    logit_error  rf_error  cox_error
5     -0.002840  0.000247  -0.002491
11     0.000217  0.000928   0.000668
17     0.011353  0.010479   0.010905
23     0.034230  0.034591   0.035673
```

16 Variable importance

[188]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from copy import deepcopy

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
```

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.calibration import CalibratedClassifierCV
from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    brier_score_loss,
    log_loss,
)
from sklearn.inspection import permutation_importance
```

[189]:
```python
target = "hazard_target"

all_numeric_features = [
    "loan_amnt",
    "term_num",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "fico_x_mob",
    "term_x_mob",
    "dti_x_mob",
]

all_categorical_features = [
    "emp_length",
    "home_ownership",
    "verification_status",
    "purpose",
    "addr_state",
    "mob_str",
    "mob_6bin",
]

# Group variables in a way that matches your reasoning
feature_groups = {
```

```
    "loan_size_terms": ["loan_amnt", "term_num", "installment"],
    "employment_home": ["emp_length", "home_ownership"],
    "income_verification": ["annual_inc", "verification_status"],
    "purpose_state": ["purpose", "addr_state"],
    "delinq_inquiry": ["delinq_2yrs", "inq_last_6mths"],
    "accounts_public": ["open_acc", "pub_rec", "pub_rec_bankruptcies"],
    "utilization_balance": ["revol_bal", "revol_util"],
    "capacity_depth": ["total_acc", "mort_acc", "credit_history_years"],
    "credit_quality": ["fico_avg", "dti"],
    "time_main": ["mob_str", "mob_6bin"],
    "time_interactions": ["fico_x_mob", "term_x_mob", "dti_x_mob"],
}

# Also test single raw features if you want finer simplification
single_feature_groups = {f: [f] for f in (all_numeric_features +␣
 ↪all_categorical_features)}
```

[190]:
```
def split_feature_types(features, numeric_pool, categorical_pool):
    num = [f for f in features if f in numeric_pool]
    cat = [f for f in features if f in categorical_pool]
    return num, cat


def make_preprocessor(numeric_features, categorical_features):
    numeric_transformer = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
    ])

    categorical_transformer = Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore")),
    ])

    preprocessor = ColumnTransformer([
        ("num", numeric_transformer, numeric_features),
        ("cat", categorical_transformer, categorical_features),
    ])
    return preprocessor


def make_logit_pipeline(numeric_features, categorical_features):
    preprocessor = make_preprocessor(numeric_features, categorical_features)
    model = Pipeline([
        ("prep", preprocessor),
        ("model", LogisticRegression(max_iter=2000))
    ])
```

```python
    return model


def make_rf_pipeline(numeric_features, categorical_features, random_state=42):
    preprocessor = make_preprocessor(numeric_features, categorical_features)

    rf_base = Pipeline([
        ("prep", preprocessor),
        ("model", RandomForestClassifier(
            n_estimators=200,
            min_samples_leaf=50,
            random_state=random_state,
            n_jobs=-1,
            class_weight="balanced_subsample"
        ))
    ])

    rf_calibrated = CalibratedClassifierCV(
        estimator=rf_base,
        method="sigmoid",
        cv=3
    )
    return rf_calibrated


def get_pred_proba(model, X):
    return model.predict_proba(X)[:, 1]


def compute_metrics(y_true, y_prob, eps=1e-12):
    y_prob = np.clip(y_prob, eps, 1 - eps)
    return {
        "roc_auc": roc_auc_score(y_true, y_prob),
        "pr_auc": average_precision_score(y_true, y_prob),
        "brier": brier_score_loss(y_true, y_prob),
        "logloss": log_loss(y_true, y_prob),
        "mean_pred": float(np.mean(y_prob)),
        "event_rate": float(np.mean(y_true)),
    }


def fit_and_score_model(
    model_type,
    features,
    X_train_full, y_train,
    X_valid_full, y_valid,
    X_test_full, y_test,
```

```
    numeric_pool,
    categorical_pool,
    random_state=42
):
    numeric_features, categorical_features = split_feature_types(
        features, numeric_pool, categorical_pool
    )

    X_train = X_train_full[features].copy()
    X_valid = X_valid_full[features].copy()
    X_test = X_test_full[features].copy()

    if model_type == "logit":
        model = make_logit_pipeline(numeric_features, categorical_features)
    elif model_type == "rf":
        model = make_rf_pipeline(numeric_features, categorical_features,␣
 ↪random_state=random_state)
    else:
        raise ValueError("model_type must be 'logit' or 'rf'")

    model.fit(X_train, y_train)

    valid_prob = get_pred_proba(model, X_valid)
    test_prob = get_pred_proba(model, X_test)

    valid_metrics = compute_metrics(y_valid, valid_prob)
    test_metrics = compute_metrics(y_test, test_prob)

    out = {
        "model": model,
        "features": features,
        "n_features_raw": len(features),
        "numeric_features": numeric_features,
        "categorical_features": categorical_features,
        "valid_metrics": valid_metrics,
        "test_metrics": test_metrics,
        "valid_prob": valid_prob,
        "test_prob": test_prob,
    }
    return out
```

[191]:
```
all_features = all_numeric_features + all_categorical_features

X_train_full = train_panel[all_features].copy()
y_train = train_panel[target].copy()

X_valid_full = valid_panel[all_features].copy()
```

```
y_valid = valid_panel[target].copy()

X_test_full = test_panel[all_features].copy()
y_test = test_panel[target].copy()
```

[192]:
```
baseline_logit = fit_and_score_model(
    model_type="logit",
    features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

baseline_rf = fit_and_score_model(
    model_type="rf",
    features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

print("Baseline logit VALID:", baseline_logit["valid_metrics"])
print("Baseline logit TEST :", baseline_logit["test_metrics"])
print()
print("Baseline RF VALID   :", baseline_rf["valid_metrics"])
print("Baseline RF TEST    :", baseline_rf["test_metrics"])
```

```
Baseline logit VALID: {'roc_auc': 0.6644510998975379, 'pr_auc':
0.013513873187659108, 'brier': 0.006965209814027969, 'logloss':
0.0407007804409802, 'mean_pred': 0.0062547019512976, 'event_rate':
0.007015858497339033}
Baseline logit TEST : {'roc_auc': 0.6705443367805358, 'pr_auc':
0.008341943712689478, 'brier': 0.004787181172564742, 'logloss':
0.03100861445099916, 'mean_pred': 0.005523270804034634, 'event_rate':
0.004435481108093726}

Baseline RF VALID   : {'roc_auc': 0.6568373154296456, 'pr_auc':
0.012907198473836468, 'brier': 0.006951580357783106, 'logloss':
0.040753960036154176, 'mean_pred': 0.00624411401652934, 'event_rate':
0.007015858497339033}
Baseline RF TEST    : {'roc_auc': 0.6652843789467573, 'pr_auc':
0.008311529282920003, 'brier': 0.004409581096031979, 'logloss':
0.027774702969589348, 'mean_pred': 0.0051076744521263056, 'event_rate':
0.004435481108093726}
```

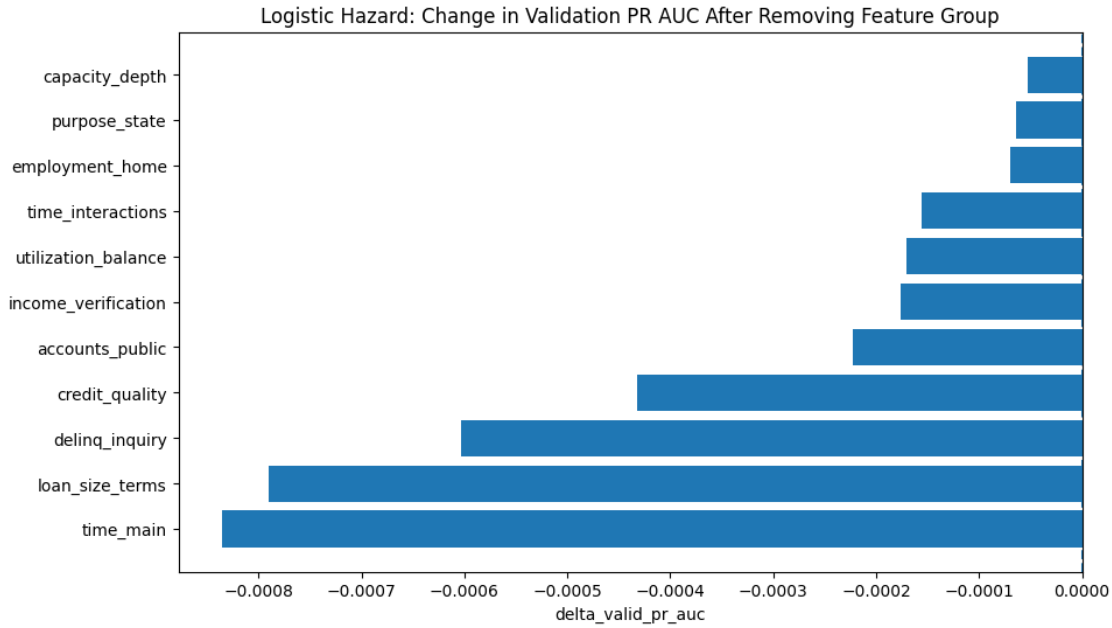
[193]:
```python
def run_group_removal_experiments(
    model_type,
    feature_groups,
    base_features,
    X_train_full, y_train,
    X_valid_full, y_valid,
    X_test_full, y_test,
    numeric_pool,
    categorical_pool,
    random_state=42
):
    baseline = fit_and_score_model(
        model_type=model_type,
        features=base_features,
        X_train_full=X_train_full, y_train=y_train,
        X_valid_full=X_valid_full, y_valid=y_valid,
        X_test_full=X_test_full, y_test=y_test,
        numeric_pool=numeric_pool,
        categorical_pool=categorical_pool,
        random_state=random_state
    )

    rows = []

    for group_name, remove_vars in feature_groups.items():
        reduced_features = [f for f in base_features if f not in remove_vars]

        result = fit_and_score_model(
            model_type=model_type,
            features=reduced_features,
            X_train_full=X_train_full, y_train=y_train,
            X_valid_full=X_valid_full, y_valid=y_valid,
            X_test_full=X_test_full, y_test=y_test,
            numeric_pool=numeric_pool,
            categorical_pool=categorical_pool,
            random_state=random_state
        )

        row = {
            "model_type": model_type,
            "removed_group": group_name,
            "removed_vars": ", ".join(remove_vars),
            "n_removed_raw": len(remove_vars),
            "n_features_remaining_raw": len(reduced_features),

            "baseline_valid_roc_auc": baseline["valid_metrics"]["roc_auc"],
            "baseline_valid_pr_auc": baseline["valid_metrics"]["pr_auc"],
```

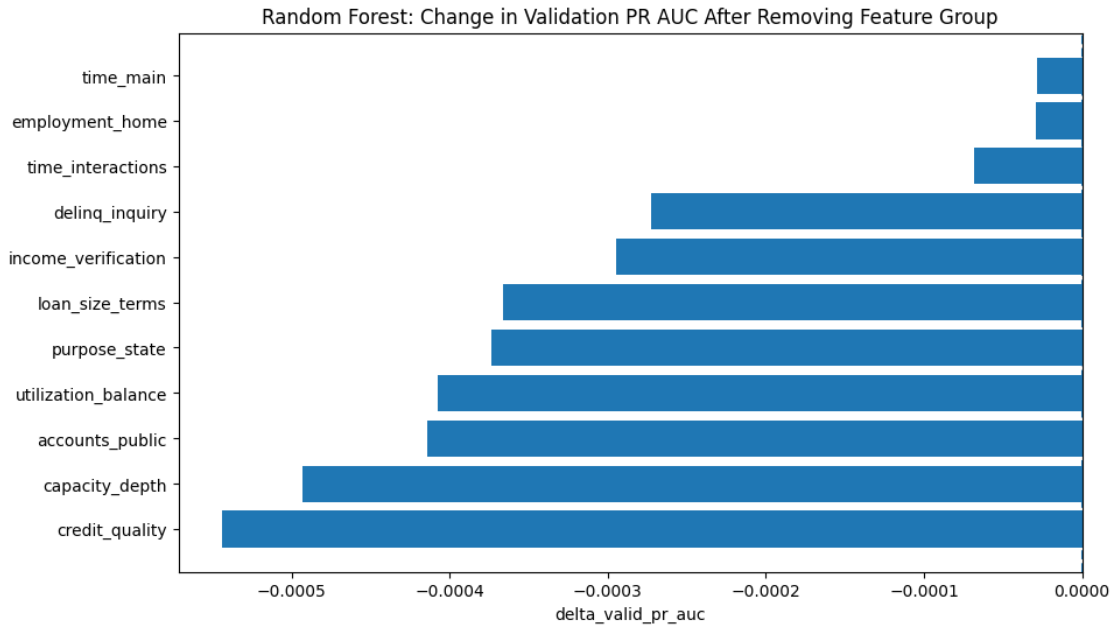
```
            "baseline_valid_brier": baseline["valid_metrics"]["brier"],
            "baseline_valid_logloss": baseline["valid_metrics"]["logloss"],

            "reduced_valid_roc_auc": result["valid_metrics"]["roc_auc"],
            "reduced_valid_pr_auc": result["valid_metrics"]["pr_auc"],
            "reduced_valid_brier": result["valid_metrics"]["brier"],
            "reduced_valid_logloss": result["valid_metrics"]["logloss"],

            "delta_valid_roc_auc": result["valid_metrics"]["roc_auc"] -␣
↪baseline["valid_metrics"]["roc_auc"],
            "delta_valid_pr_auc": result["valid_metrics"]["pr_auc"] -␣
↪baseline["valid_metrics"]["pr_auc"],
            "delta_valid_brier": result["valid_metrics"]["brier"] -␣
↪baseline["valid_metrics"]["brier"],
            "delta_valid_logloss": result["valid_metrics"]["logloss"] -␣
↪baseline["valid_metrics"]["logloss"],

            "baseline_test_roc_auc": baseline["test_metrics"]["roc_auc"],
            "baseline_test_pr_auc": baseline["test_metrics"]["pr_auc"],
            "baseline_test_brier": baseline["test_metrics"]["brier"],
            "baseline_test_logloss": baseline["test_metrics"]["logloss"],

            "reduced_test_roc_auc": result["test_metrics"]["roc_auc"],
            "reduced_test_pr_auc": result["test_metrics"]["pr_auc"],
            "reduced_test_brier": result["test_metrics"]["brier"],
            "reduced_test_logloss": result["test_metrics"]["logloss"],

            "delta_test_roc_auc": result["test_metrics"]["roc_auc"] -␣
↪baseline["test_metrics"]["roc_auc"],
            "delta_test_pr_auc": result["test_metrics"]["pr_auc"] -␣
↪baseline["test_metrics"]["pr_auc"],
            "delta_test_brier": result["test_metrics"]["brier"] -␣
↪baseline["test_metrics"]["brier"],
            "delta_test_logloss": result["test_metrics"]["logloss"] -␣
↪baseline["test_metrics"]["logloss"],
        }
        rows.append(row)

    results_df = pd.DataFrame(rows).sort_values(
        ["delta_valid_pr_auc", "delta_valid_roc_auc"],
        ascending=[False, False]
    ).reset_index(drop=True)

    return baseline, results_df
```

[ ]:
```
baseline_logit_grouped, grouped_logit_results = run_group_removal_experiments(
    model_type="logit",
    feature_groups=feature_groups,
    base_features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

baseline_rf_grouped, grouped_rf_results = run_group_removal_experiments(
    model_type="rf",
    feature_groups=feature_groups,
    base_features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

grouped_logit_results
```

[ ]:
```
def plot_group_removal(results_df, metric_col, title):
    plot_df = results_df.sort_values(metric_col, ascending=True)

    plt.figure(figsize=(10, 6))
    plt.barh(plot_df["removed_group"], plot_df[metric_col])
    plt.axvline(0, linestyle="--")
    plt.xlabel(metric_col)
    plt.title(title)
    plt.show()
```

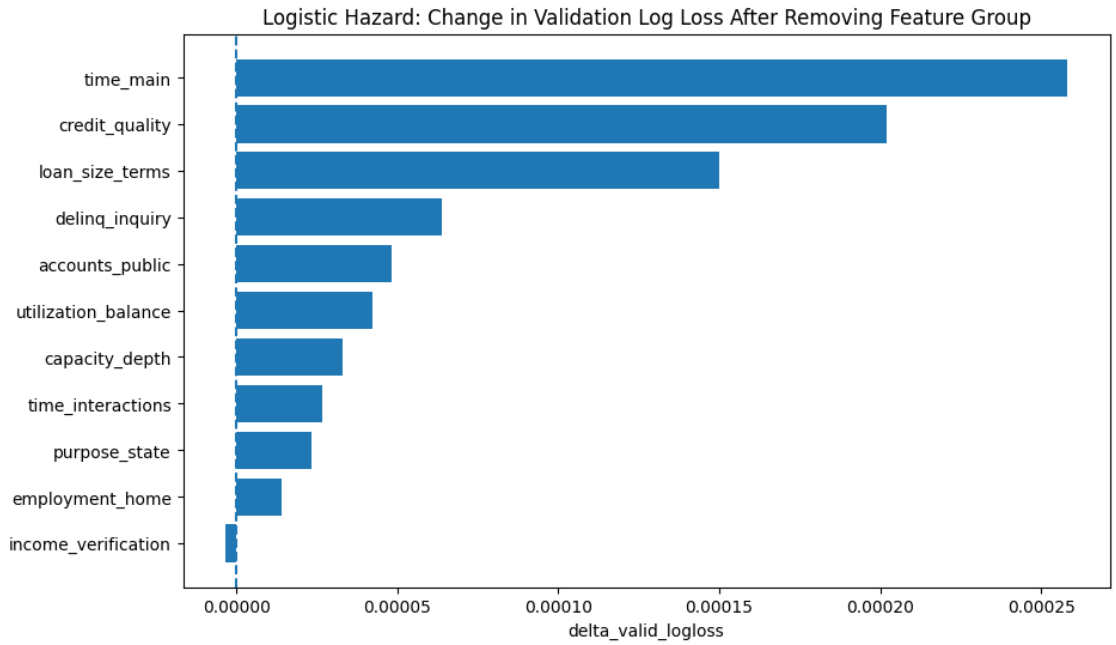
[ ]:
```
plot_group_removal(
    grouped_logit_results,
    "delta_valid_pr_auc",
    "Logistic Hazard: Change in Validation PR AUC After Removing Feature Group"
)

plot_group_removal(
    grouped_rf_results,
    "delta_valid_pr_auc",
    "Random Forest: Change in Validation PR AUC After Removing Feature Group"
)
```

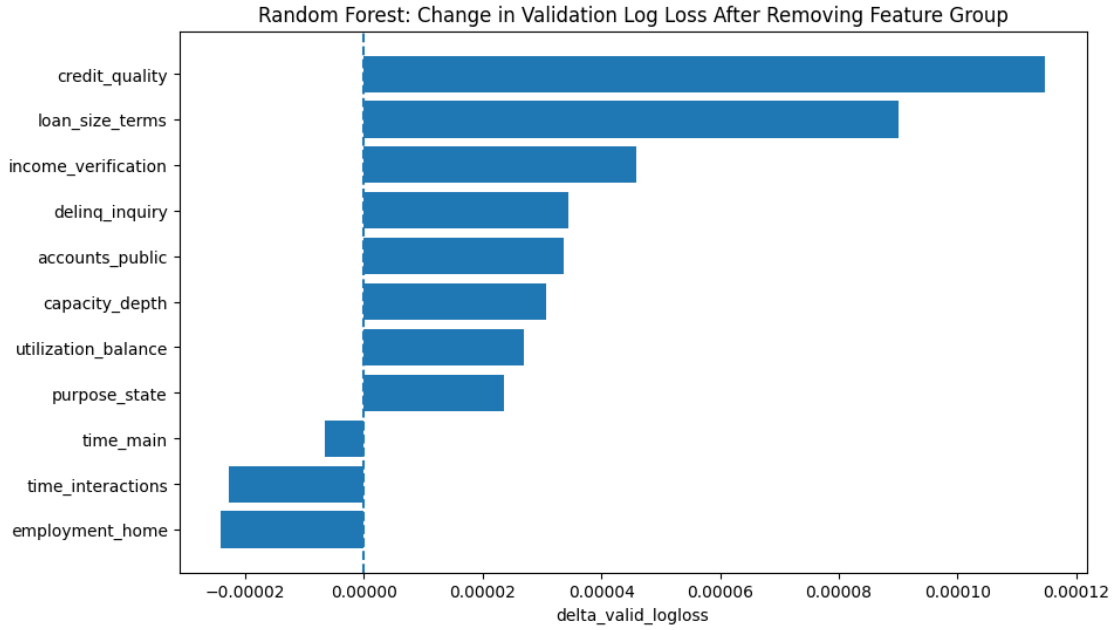




[ ]:
```
plot_group_removal(
    grouped_logit_results,
    "delta_valid_logloss",
    "Logistic Hazard: Change in Validation Log Loss After Removing Feature␣
↪Group"
```

```
)

plot_group_removal(
    grouped_rf_results,
    "delta_valid_logloss",
    "Random Forest: Change in Validation Log Loss After Removing Feature Group"
)
```





[ ]:
```
baseline_logit_single, single_logit_results = run_group_removal_experiments(
    model_type="logit",
    feature_groups=single_feature_groups,
    base_features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

baseline_rf_single, single_rf_results = run_group_removal_experiments(
    model_type="rf",
    feature_groups=single_feature_groups,
    base_features=all_features,
    X_train_full=X_train_full, y_train=y_train,
    X_valid_full=X_valid_full, y_valid=y_valid,
    X_test_full=X_test_full, y_test=y_test,
    numeric_pool=all_numeric_features,
    categorical_pool=all_categorical_features,
)

single_logit_results.head(20)
```

[ ]:
```
   model_type         removed_group          removed_vars  n_removed_raw  \
0       logit              mort_acc              mort_acc              1
1       logit               pub_rec               pub_rec              1
2       logit           delinq_2yrs           delinq_2yrs              1
3       logit              open_acc              open_acc              1
4       logit            emp_length            emp_length              1
5       logit            addr_state            addr_state              1
6       logit            revol_util            revol_util              1
7       logit               purpose               purpose              1
8       logit            annual_inc            annual_inc              1
9       logit              mob_6bin              mob_6bin              1
10      logit             total_acc             total_acc              1
11      logit            term_x_mob            term_x_mob              1
12      logit   verification_status   verification_status              1
13      logit               mob_str               mob_str              1
14      logit  pub_rec_bankruptcies  pub_rec_bankruptcies              1
15      logit  credit_history_years  credit_history_years              1
16      logit             revol_bal             revol_bal              1
17      logit            fico_x_mob            fico_x_mob              1
18      logit             loan_amnt             loan_amnt              1
```

19	logit	fico_avg	fico_avg	1

	n_features_remaining_raw	baseline_valid_roc_auc	baseline_valid_pr_auc	\
0	25	0.668269	0.013058	
1	25	0.668269	0.013058	
2	25	0.668269	0.013058	
3	25	0.668269	0.013058	
4	25	0.668269	0.013058	
5	25	0.668269	0.013058	
6	25	0.668269	0.013058	
7	25	0.668269	0.013058	
8	25	0.668269	0.013058	
9	25	0.668269	0.013058	
10	25	0.668269	0.013058	
11	25	0.668269	0.013058	
12	25	0.668269	0.013058	
13	25	0.668269	0.013058	
14	25	0.668269	0.013058	
15	25	0.668269	0.013058	
16	25	0.668269	0.013058	
17	25	0.668269	0.013058	
18	25	0.668269	0.013058	
19	25	0.668269	0.013058	

	baseline_valid_brier	baseline_valid_logloss	reduced_valid_roc_auc	\
0	0.006849	0.040077	0.669209	
1	0.006849	0.040077	0.668363	
2	0.006849	0.040077	0.668246	
3	0.006849	0.040077	0.669468	
4	0.006849	0.040077	0.669764	
5	0.006849	0.040077	0.668599	
6	0.006849	0.040077	0.667980	
7	0.006849	0.040077	0.666070	
8	0.006849	0.040077	0.668251	
9	0.006849	0.040077	0.667141	
10	0.006849	0.040077	0.665963	
11	0.006849	0.040077	0.666285	
12	0.006849	0.040077	0.666446	
13	0.006849	0.040077	0.664234	
14	0.006849	0.040077	0.667815	
15	0.006849	0.040077	0.667714	
16	0.006849	0.040077	0.665202	
17	0.006849	0.040077	0.665285	
18	0.006849	0.040077	0.665474	
19	0.006849	0.040077	0.660921	

	reduced_valid_pr_auc	reduced_valid_brier	reduced_valid_logloss	\

0	0.013136	0.006849	0.040072
1	0.013084	0.006849	0.040073
2	0.013080	0.006849	0.040074
3	0.013046	0.006849	0.040061
4	0.013043	0.006849	0.040073
5	0.013005	0.006849	0.040074
6	0.012989	0.006849	0.040075
7	0.012957	0.006850	0.040108
8	0.012921	0.006850	0.040047
9	0.012920	0.006850	0.040103
10	0.012888	0.006850	0.040112
11	0.012863	0.006850	0.040109
12	0.012860	0.006850	0.040116
13	0.012840	0.006850	0.040147
14	0.012830	0.006850	0.040092
15	0.012830	0.006850	0.040092
16	0.012829	0.006850	0.040126
17	0.012828	0.006850	0.040118
18	0.012827	0.006850	0.040118
19	0.012809	0.006851	0.040219

	delta_valid_roc_auc	delta_valid_pr_auc	delta_valid_brier	\
0	0.000940	0.000078	-7.725289e-08	
1	0.000094	0.000027	-9.519762e-08	
2	-0.000023	0.000023	-4.181435e-08	
3	0.001199	-0.000011	-9.749979e-08	
4	0.001495	-0.000015	6.565881e-08	
5	0.000330	-0.000052	7.532725e-08	
6	-0.000289	-0.000068	-1.087121e-08	
7	-0.002199	-0.000101	4.736489e-07	
8	-0.000018	-0.000137	3.115668e-07	
9	-0.001128	-0.000137	5.524725e-07	
10	-0.002306	-0.000170	5.979583e-07	
11	-0.001984	-0.000194	5.752463e-07	
12	-0.001823	-0.000198	7.089803e-07	
13	-0.004035	-0.000217	9.320598e-07	
14	-0.000454	-0.000227	6.201298e-07	
15	-0.000555	-0.000227	6.622002e-07	
16	-0.003067	-0.000229	7.861827e-07	
17	-0.002984	-0.000230	6.968935e-07	
18	-0.002795	-0.000231	7.023082e-07	
19	-0.007348	-0.000249	2.083073e-06	

	delta_valid_logloss	baseline_test_roc_auc	baseline_test_pr_auc	\
0	-0.000005	0.668531	0.010645	
1	-0.000004	0.668531	0.010645	
2	-0.000002	0.668531	0.010645	

3	-0.000015	0.668531	0.010645
4	-0.000004	0.668531	0.010645
5	-0.000003	0.668531	0.010645
6	-0.000002	0.668531	0.010645
7	0.000031	0.668531	0.010645
8	-0.000030	0.668531	0.010645
9	0.000026	0.668531	0.010645
10	0.000036	0.668531	0.010645
11	0.000032	0.668531	0.010645
12	0.000039	0.668531	0.010645
13	0.000070	0.668531	0.010645
14	0.000015	0.668531	0.010645
15	0.000015	0.668531	0.010645
16	0.000049	0.668531	0.010645
17	0.000041	0.668531	0.010645
18	0.000042	0.668531	0.010645
19	0.000143	0.668531	0.010645

	baseline_test_brier	baseline_test_logloss	reduced_test_roc_auc	\
0	0.005765	0.035015	0.669944	
1	0.005765	0.035015	0.669152	
2	0.005765	0.035015	0.668670	
3	0.005765	0.035015	0.673609	
4	0.005765	0.035015	0.673713	
5	0.005765	0.035015	0.667389	
6	0.005765	0.035015	0.668818	
7	0.005765	0.035015	0.666435	
8	0.005765	0.035015	0.670144	
9	0.005765	0.035015	0.673646	
10	0.005765	0.035015	0.664363	
11	0.005765	0.035015	0.664910	
12	0.005765	0.035015	0.670584	
13	0.005765	0.035015	0.663190	
14	0.005765	0.035015	0.674079	
15	0.005765	0.035015	0.673260	
16	0.005765	0.035015	0.664659	
17	0.005765	0.035015	0.663379	
18	0.005765	0.035015	0.663529	
19	0.005765	0.035015	0.657797	

	reduced_test_pr_auc	reduced_test_brier	reduced_test_logloss	\
0	0.010713	0.005770	0.035071	
1	0.010679	0.005766	0.035014	
2	0.010666	0.005764	0.034995	
3	0.010828	0.005776	0.035105	
4	0.010841	0.005759	0.034849	
5	0.010491	0.005760	0.034962	

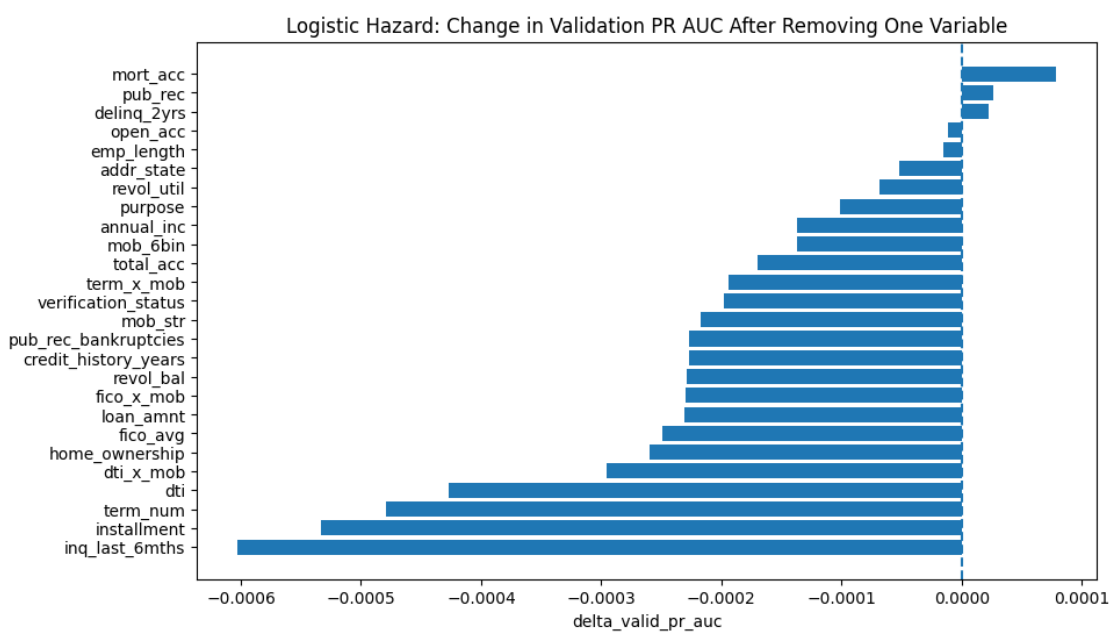
6	0.010647	0.005766	0.035022
7	0.010632	0.005773	0.035142
8	0.010636	0.005827	0.035794
9	0.010877	0.005755	0.034802
10	0.010436	0.005747	0.034760
11	0.010430	0.005765	0.035034
12	0.010654	0.005766	0.034997
13	0.010442	0.005734	0.034607
14	0.010720	0.005760	0.034857
15	0.010755	0.005763	0.034911
16	0.010412	0.005751	0.034815
17	0.010405	0.005744	0.034741
18	0.010380	0.005748	0.034789
19	0.010299	0.005734	0.034711

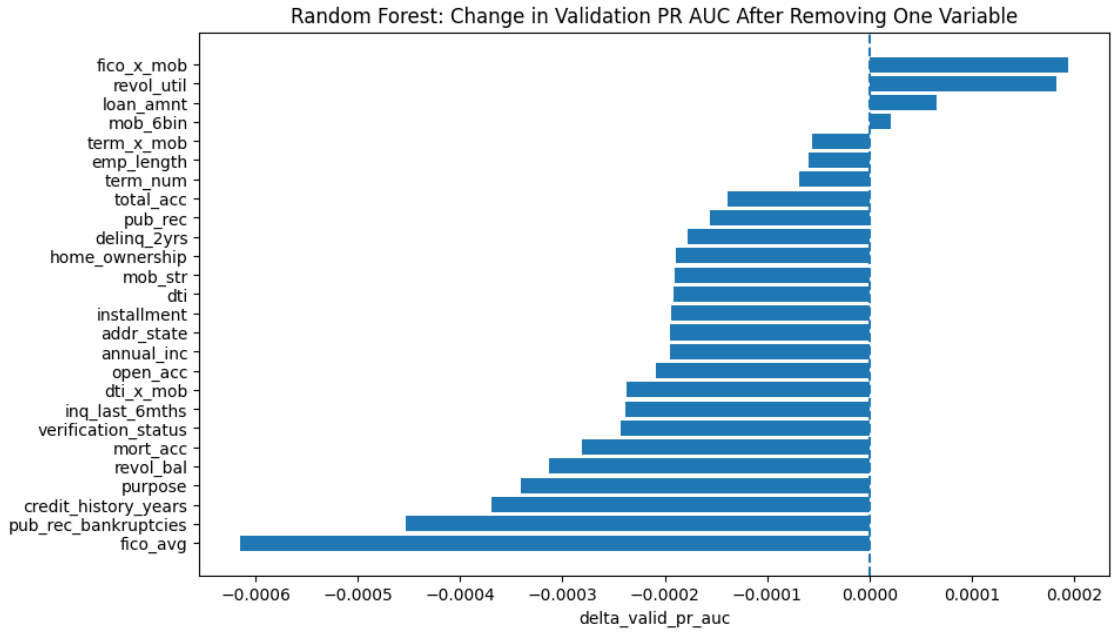
	delta_test_roc_auc	delta_test_pr_auc	delta_test_brier \
0	0.001414	0.000068	4.905618e-06
1	0.000622	0.000034	3.745275e-07
2	0.000139	0.000021	-7.654378e-07
3	0.005078	0.000183	1.131944e-05
4	0.005182	0.000196	-6.158057e-06
5	-0.001141	-0.000153	-5.272798e-06
6	0.000287	0.000002	1.162240e-06
7	-0.002095	-0.000013	7.887955e-06
8	0.001613	-0.000009	6.215022e-05
9	0.005115	0.000232	-9.769784e-06
10	-0.004168	-0.000209	-1.854577e-05
11	-0.003620	-0.000215	-2.416350e-07
12	0.002053	0.000010	8.611655e-07
13	-0.005341	-0.000203	-3.085581e-05
14	0.005549	0.000075	-4.933083e-06
15	0.004730	0.000111	-1.834553e-06
16	-0.003871	-0.000233	-1.439417e-05
17	-0.005151	-0.000239	-2.147233e-05
18	-0.005001	-0.000265	-1.697250e-05
19	-0.010734	-0.000346	-3.116763e-05

	delta_test_logloss
0	5.599137e-05
1	-3.471072e-07
2	-1.954630e-05
3	9.016541e-05
4	-1.656655e-04
5	-5.277154e-05
6	7.671090e-06
7	1.268175e-04
8	7.788766e-04

```
9        -2.129020e-04
10       -2.552166e-04
11        1.871241e-05
12       -1.786710e-05
13       -4.076950e-04
14       -1.575671e-04
15       -1.034938e-04
16       -2.000891e-04
17       -2.740516e-04
18       -2.256541e-04
19       -3.039194e-04
```

[ ]:
```
plot_group_removal(
    single_logit_results,
    "delta_valid_pr_auc",
    "Logistic Hazard: Change in Validation PR AUC After Removing One Variable"
)

plot_group_removal(
    single_rf_results,
    "delta_valid_pr_auc",
    "Random Forest: Change in Validation PR AUC After Removing One Variable"
)
```





[ ]:
```python
def flag_removal_candidates(results_df, pr_tol=0.002, logloss_tol=0.005):
    out = results_df.copy()

    out["safe_to_remove"] = (
        (out["delta_valid_pr_auc"] >= -pr_tol) &
        (out["delta_valid_logloss"] <= logloss_tol)
    )

    return out.sort_values(
        ["safe_to_remove", "delta_valid_pr_auc", "delta_valid_logloss"],
        ascending=[False, False, True]
    )

flagged_logit = flag_removal_candidates(grouped_logit_results)
flagged_rf = flag_removal_candidates(grouped_rf_results)

flagged_logit[[
    "removed_group", "removed_vars",
    "delta_valid_pr_auc", "delta_valid_roc_auc",
    "delta_valid_logloss", "safe_to_remove"
]]
```

[ ]:
```
          removed_group                                removed_vars  \
0        capacity_depth  total_acc, mort_acc, credit_history_years
1         purpose_state                         purpose, addr_state
2       employment_home                  emp_length, home_ownership
```

```
3     time_interactions          fico_x_mob, term_x_mob, dti_x_mob
4   utilization_balance                      revol_bal, revol_util
5   income_verification            annual_inc, verification_status
6       accounts_public    open_acc, pub_rec, pub_rec_bankruptcies
7        credit_quality                              fico_avg, dti
8        delinq_inquiry                delinq_2yrs, inq_last_6mths
9       loan_size_terms           loan_amnt, term_num, installment
10            time_main                          mob_str, mob_6bin

    delta_valid_pr_auc  delta_valid_roc_auc  delta_valid_logloss  \
0            -0.000052            -0.000946             0.000033
1            -0.000064            -0.002146             0.000023
2            -0.000070            -0.000404             0.000014
3            -0.000156            -0.001340             0.000027
4            -0.000171            -0.002192             0.000042
5            -0.000176            -0.002078            -0.000004
6            -0.000223            -0.002675             0.000048
7            -0.000433            -0.010802             0.000202
8            -0.000603            -0.004724             0.000064
9            -0.000791            -0.010124             0.000150
10           -0.000836            -0.016030             0.000258

    safe_to_remove
0             True
1             True
2             True
3             True
4             True
5             True
6             True
7             True
8             True
9             True
10            True
```

[ ]:
```
groups_to_remove_logit = flagged_logit.loc[
    flagged_logit["safe_to_remove"], "removed_group"
].tolist()

groups_to_remove_logit
```

[ ]:
```
['capacity_depth',
 'purpose_state',
 'employment_home',
 'time_interactions',
 'utilization_balance',
 'income_verification',
```

```
'accounts_public',
'credit_quality',
'delinq_inquiry',
'loan_size_terms',
'time_main']
```